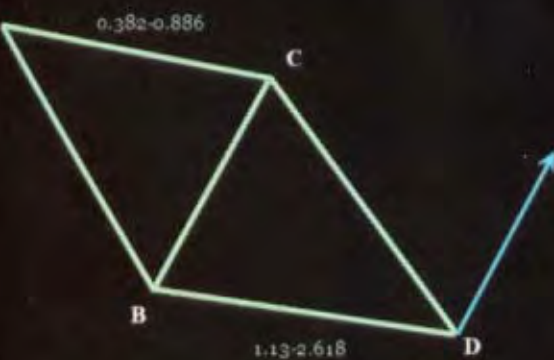
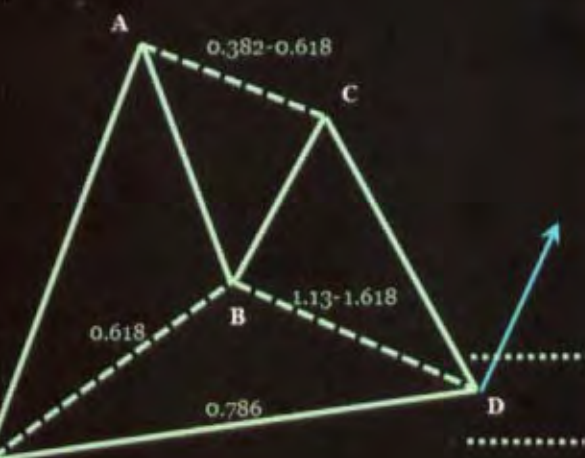




ABCD (Bullish)



Gartley (Bullish)



2B Buy (Bullish)

Trade Chart Patterns Like The Pros

Specific Trading Techniques

SURI DUDDELLA

Reviews for Suri Duddella's Trade Chart Patterns Like The Pros

"Don't read this book – MAKE MONEY WITH IT! *Trade Chart Patterns Like The Pros* by Suri Duddella is one of those rare opportunities to build a base of knowledge that can take you further each time you trade. Suri has laid out an indispensable tome on trading chart patterns based upon his real-life experiences in a format that will work well for the spectrum of investors – from novice to expert. *Trade Chart Patterns Like The Pros* stays focused and does not overwhelm you with mathematical statistics and technical jargon that will surely turn you off."

"... Suri's "trader-sense" perspective helps you identify a pattern and apply the relevant techniques to enter, manage and exit the trade. Each of his 65 patterns includes a brief synopsis written in plain English and an actual chart to reinforce the concept, not a conveniently drawn perfect example that never occurs in real life. Suri explains the setup and then provides the entry point triggers as well as an exit strategy with targets for profitable trades and stops to minimize any losses."

"...Suri's practical approach will give you insights that will make it harder to lose. Great investing books never sit on the bookshelf – they stay at your side as a trading partner and I can think of no better trader to have at my side than Suri Duddella. *Trade Chart Patterns Like The Pros* is destined to become one of those rare TA books that you spend less time reading and more time using to MAKE MONEY."

--- Michael Steinhardt, HEDGEfolios founder

"...Anyone who has had the privilege of trading alongside him will tell you that Suri's trading skills are second to none. He is consistent and disciplined, but what sets Suri apart is his ability to find and exploit regular market patterns.

This book shares Suri's techniques in detail. For the past two years, I have watched the book develop from a concept into a finished product. The results have exceeded even my high expectations, and I will certainly keep a dog-eared copy within arm's reach of my own trading turret. This book answers the "how" question. It reveals actual techniques that top traders actually can and do use, in a format that lets you use them yourself.

Beginning and experienced traders alike will benefit from this book. Applying these techniques can help the rest of us improve our trading too. "

--- Joshua Silverman

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Trading is risky. You are 100% responsible for your own trading. The author, Suri Duddella, specifically disclaims any and all express and implied warranties. Your trades may entail substantial loss. Nothing in this book should be construed as a recommendation to buy or sell any security or other instrument, or a determination that any trade is suitable for you.

The examples in this book could be considered hypothetical trades. The CFTC warns that:

HYPOTHETICAL PERFORMANCE RESULTS HAVE MANY INHERENT LIMITATIONS, SOME OF WHICH ARE DESCRIBED BELOW. NO REPRESENTATION IS BEING MADE THAT ANY ACCOUNT WILL OR IS LIKELY TO ACHIEVE PROFITS OR LOSSES SIMILAR TO THOSE SHOWN. IN FACT, THERE ARE FREQUENTLY SHARP DIFFERENCES BETWEEN HYPOTHETICAL PERFORMANCE RESULTS AND THE ACTUAL RESULTS SUBSEQUENTLY ACHIEVED BY ANY PARTICULAR TRADING PROGRAM. ONE OF THE LIMITATIONS OF HYPOTHETICAL PERFORMANCE RESULTS IS THAT THEY ARE GENERALLY PREPARED WITH THE BENEFIT OF HINDSIGHT. IN ADDITION, HYPOTHETICAL TRADING DOES NOT INVOLVE FINANCIAL RISK, AND NO HYPOTHETICAL TRADING RECORD CAN COMPLETELY ACCOUNT FOR THE IMPACT OF FINANCIAL RISK IN ACTUAL TRADING. FOR EXAMPLE, THE ABILITY TO WITHSTAND LOSSES OR TO ADHERE TO A PARTICULAR TRADING PROGRAM IN SPITE OF TRADING LOSSES ARE MATERIAL POINTS WHICH CAN ALSO ADVERSELY AFFECT ACTUAL TRADING RESULTS. THERE ARE NUMEROUS OTHER FACTORS RELATED TO THE MARKETS IN GENERAL OR TO THE IMPLEMENTATION OF ANY SPECIFIC TRADING PROGRAM WHICH CANNOT BE FULLY ACCOUNTED FOR IN THE PREPARATION OF HYPOTHETICAL PERFORMANCE RESULTS AND ALL OF WHICH CAN ADVERSELY AFFECT ACTUAL TRADING RESULTS.

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ACKNOWLEDGMENTS

It has been many years since I first thought of writing this book. Sometimes I was discouraged about finding the time necessary to write such a book. Sometimes a clear concept or design for the book was mentally unavailable. With encouragement and support from some of my friends, I finally gained the courage, time, motivation and focus to begin the project. Below are the friends who encouraged me to write this book and who have significantly influenced my ideas and concepts. My sincere thanks to them all. Without their help, this book would not have been possible.

Dan (“Dr”) Doss
Josh Silverman
Michael Steinhardt
Suneeta Bindal
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Mark Booher
Michael Taylor
TradeStation Securities

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Also, I sincerely thank Scott Carney of harmonictrader.com and Michael Jardine of enthios.com who have graciously given me permission to use a portion of their work on Harmonic patterns and Market Structures in this book.

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FOREWORD by Joshua Silverman

There is a famous basketball court in New York City called Rucker Park. Legend has it that the amateurs who played there were so good, they could take on the pros. There were no shoe contracts or television cameras, but fans sitting on those worn bleachers could see some of the best pure basketball anywhere.

If there were a Rucker Park for trading, Suri Duddella would hold court there. He may not be a regular guest on CNBC or run a billion dollar hedge fund, but anyone who has had the privilege of trading alongside him will tell you that Suri's trading skills are second to none. He is consistent and disciplined, but what sets Suri apart is his ability to find and exploit regular market patterns.

Traders at that level rarely let readers have more than a peek behind the curtain. This book shares Suri's techniques in detail. For the past two years, I have watched the book develop from a concept into a finished product. The results have exceeded even my high expectations, and I will certainly keep a dog-eared copy within arm's reach of my own trading turret.

Plenty of books will tell you what a flag or diamond or Gartley pattern is. Many will tell you whether the patterns are bullish or bearish. But that's where the detail stops. Because most of these books are written by professional authors, not traders, they cannot provide any guidance on exactly *how* to trade these patterns. This book answers the 'how' question. It reveals actual techniques that top traders actually can and do use, in a format that lets you use them yourself.

Beginning and experienced traders alike will benefit from this book. The biggest beneficiary, though, is probably Suri Duddella himself. Writing this book has forced him to define and hone his techniques. I have watched him bring his trading game to the next level. He is too modest to admit it, but even if this book doesn't sell a single copy, it has already been a huge success for him. Applying these techniques can help the rest of us improve our trading too.

Joshua Silverman
Chicago, IL

REVIEW by Michael Steinhardt

Don't read this book – MAKE MONEY WITH IT! There are hundreds of Technical Analysis books and only a handful will ever help you make enough successful trades to recover the purchase price. *Trade Chart Patterns Like The Pros* by Suri Duddella is one of those rare opportunities to build a base of knowledge that can take you further each time you trade.

Like the Chinese proverb that says, "Give a man a fish and you feed him for a day. Teach a man to fish and you feed him for a lifetime." Suri has laid out an indispensable tome on trading chart patterns based upon his real-life experiences in a format that will work well for the spectrum of investors – from novice to expert. Unlike the 'encyclopedias' you can find on this unbelievably deep and complex subject, *Trade Chart Patterns Like The Pros* stays focused and does not overwhelm you with mathematical statistics and technical jargon that will surely turn you off.

Instead, Suri's 'trader-sense' perspective helps you identify a pattern and apply the relevant techniques to enter, manage and exit the trade. Each of his 65 patterns includes a brief synopsis written in plain English and an actual chart to reinforce the concept, not a conveniently drawn perfect example that never occurs in real life. This book provides a universal resource for 65 of the most common scenarios that you will run into regardless of your investing time frame (intraday to weeks at a time) and your preferred financial instrument. It's not enough to recognize a pattern; you need to understand the key elements of the trade and how to make it work for 'long' and 'short' positions. Suri explains the setup and then provides the entry point triggers as well as an exit strategy with targets for profitable trades and stops to minimize any losses.

What I can promise you is that Suri's practical approach will give you insights that will make it harder to lose. Great investing books never sit on the bookshelf – they stay at your side as a trading partner and I can think of no better trader to have at my side than Suri Duddella. *Trade Chart Patterns Like The Pros* is destined to become one of those rare TA books that you spend less time reading and more time using to MAKE MONEY.

Michael Steinhardt, HEDGEfolios founder

INTRODUCTION

INTRODUCTION

I have been a mathematician all my life and I am drawn to the application of mathematics in my studies and work. I studied engineering and computers, and eventually found myself involved in the financial markets. The markets offer me a never-ending challenge and the combined application of my true passions--mathematics, logic, computers and money.

When I began focusing on the financial markets in 1995, my first inclination was to study the price movements of the stocks, which in turn led me to study technical analysis. I attended every major market conference and studied many books on the markets. I was fortunate to meet some of the best market 'gurus' and learn the market ropes from them. This was the turning point for me and I decided to pursue trading as my career. Since then—for the past 12 years—I have been studying the financial markets and trading full-time rather successfully. I have made many mistakes—and still do to this day—and I am committed to learn from them. I have had the opportunity to present my trading methods to numerous conferences and have written many articles in various trade magazines. I found my niche to be Pattern Trading.

Many brilliant books have been written on Market Theories, Market Geometry and Pattern Trading. From these books I found what I consider the best strategies, trading techniques and the most reliable chart patterns. Some of the chart pattern techniques have a very high success rate. I have been trading them, and collecting and documenting the results since 1998. I considered writing a detailed reference book for my own trading and in 2005, I began to document these ideas and compile them in a book.

The best part of collecting ideas and writing about these patterns has been that it has significantly improved my own understanding of the inner workings of the markets, and in many ways has greatly improving my own trading techniques. Here is my book, *Trade Chart Patterns Like The Pros*. It provides specific and practical trading techniques and how I trade them. Each pattern is written from the perspective of the trader. Many of the techniques are my own observations and my own trading methods. Not every chart pattern will be suitable for every trader (e.g., counter trend trading, V-bottoms and V-tops may not be for you). In addition, some charts may be too elusive (e.g., diamonds) or too exotic (e.g., sea horse or dragon). Gartley and Butterfly may be too complex.

Regardless of what patterns you are interested in trading, you must master their details before trading them. Patterns must be studied thoroughly in various markets, time-frames and in all conditions. I have spent almost twelve years studying them thoroughly and they still fascinate me. I believe a few patterns are fully reliable and can be traded every time by themselves without the need of support from other indicators or confirming conditions. Other patterns need confirmation and may be less reliable traded independently. Knowing when to trade is a very important aspect of trading education and could be the 'key' to any trader's success. I constantly remind myself that technical analysis is only tool and pattern recognition is only part of the market analysis. Trading success comes from key areas like analysis, trading discipline, execution and money management skills. I will focus on Pattern Analysis and Trading in this book.

WHY THIS BOOK?

There have been many books written on market analysis, technical analysis, trading techniques and various trading methods. Most of these books detail very elaborate concepts, psychology behind the theories, what-if scenarios, statistics and excellent examples. In my view, though, many of these books miss the 'trader perspective' of when and how to trade. Therefore, in this book I will address the trading perspective rather than myriad details of pattern history, random comparisons, unproven or semi-useless statistics, or even some random stories. Most traders are anxious to know what and how to trade, and to know where the entry, the stop or the target is. This book focuses on those points and gives you details of trading every pattern. I have covered about 65 of the most unique, frequent and important patterns that are presented in day-to-day trading. I have provided credits for the original authors of some of those patterns where possible. There may be other patterns which appear frequently which I may have failed to cover. Find some patterns that pique your interest from the 65 patterns presented in this book and master them. You only need one single pattern to be successful.

NO OSCILLATORS OR INDICATORS?

This book is designed to show the patterns independently and provide trading examples of every pattern. Oscillators and momentum-based indicators have not been included. It is important that the reader fully understand just the pattern and its intra-bar relationships, and other indicators may obscure the actual understanding of these patterns. I have studied oscillators and have built excellent strategies to trade with them, but I have had much better success when I rely on solely on patterns. Hence, I have not included any oscillators in this book. I do not mean to say that oscillators and momentum-based indicators do not work—they do work with the right mind-set at the right time. Obviously, I am biased toward chart patterns and truly think they are far more reliable than momentum- or oscillator-based indicators.

WHAT ABOUT STATISTICS?

There are no perfect chart patterns and chart patterns do fail—the key is to know when to trade them and when to avoid them. So, what are the success and failure rates of the patterns? This is very valid question and will be addressed when studying the patterns.

As an engineer and mathematician I understand the power and the importance of statistics very well. To a trader, statistics must be practically useful, but too often they are not. I have seen unproven statistics presentations elsewhere, such as the failure rate waiting for a downside breakout is 32%--but if you don't wait, it is 36%. Or, the average rise is 40%, but most likely it will be 12%. Or, the percentage of high and low price breakouts subject to throwback is 33%. What do these statistics mean to a trader? Having been around the trading world the past 12 years, I know a little bit about what and how traders think. While trading, no trader remembers such statistics and no one trades using such specific statistics. Does the trader's decision change if the throwback rate is 32% instead of 33%? No, he trades what he sees, what he knows and what he has confidence in. His eyes are set on an entry, a stop and targets, and not on some unproven and semi-useless statistics. Hence, I have provided some reliability analysis about a pattern and avoided elaborate statistical analysis in this book.

BOOK STRUCTURE & FORMAT

Perspective

Most concepts in this book are compelled by a ‘trading perspective.’ After showing how a pattern has developed, the focus is on what to do next.

Format

Each Pattern section is structured to provide: a brief overview, pattern structure description and pattern trading examples with detailed entry, stop and targets.

Type of Charts

Most charts in this book are presented utilizing bar charts for illustration. I have also chosen to use some candlestick charts. All charts were created in TradeStation.

Time-Frames

Most examples are provided with tick, minute and daily charts. I have used stocks and E-mini futures for the examples and pattern descriptions.

Charting Software

TradeStation Securities

Entry Rules

In trending markets, pattern formation confirmations are critical for trading success. In counter-trend trading, there are no clear confirmations other than price trading at some level, divergence, completion of pattern structure, or plain hunch. Most of the reversal patterns need a price confirmation from price reversal zones. An important technique I implement in my trading is to find a reversal bar (the wider the better) and then entering one or two ticks above that bar's high or below the bar's low. I found significant success using this basic technique and use it in examples throughout this book. For example, if I am presenting a 'long' setup in a pattern after a trendline breakout, I initiate a 'long' trade only at 1 or 2 ticks above the breakout bar's high. Similarly, if I am presenting a short setup after a trendline breakdown, I initiate a 'short' trade only at 1 or 2 ticks below the breakdown bar's low. These entries are only valid for the next 3 to 5 bars. Beyond that, the setup may not materialize and should be avoided in most cases.

Stop Rules

Most patterns are traded after clear confirmation signals. The success of Pattern structures is dependent on market conditions. A pattern failure may occur when the pattern setup is not valid due to price-action reversals. For example, a Cup and Handle pattern formed in bear markets may fail more than in bullish markets. Even though I like dollar stops in my short-term trading, there are many times the stops are set at a critical Fibonacci confluence level, or at a major swing high or major swing low, or at some trendline support/resistance levels. Once the price breaches any of these levels, in weaker or stronger markets, the pattern structure will fail. Trading against these levels is really a futile proposition. Sometimes pattern ranges are so large that the risk-to-reward may not be attractive. Hence, stop protection at the half of the pattern range is considered. For harmonic patterns, I implement Fibonacci retracement/expansion ranges for stops.

Target Rules

Managing stops and targets is an art form. Every pattern has its own targets and stops. My trading rule is “never enter a stock just because of some price level (e.g., Fib. level, MA, Support) *alone*, but always exit because of some important price level to protect your profits.” This rule applies very well to targets. I base many of my targets on a portion (Fibonacci ratio) of the prior swing range or multiples of the prior swing range. In channel trading, it is the range of the channel or depth of the patterns, such as in head and shoulders or swing highs/swing lows prior to the pattern formations.

Fibonacci Focus

I employ Fibonacci ratios extensively to show support and resistance levels. I use a specific set of Fibonacci extensions and Fibonacci projections. The primary Fibonacci numbers I focus on are: 0.382, 0.5, 0.618, and 0.786 for the retracements and 1.272 and 1.618 for the extensions. The secondary Fibonacci focus is: 0.236, 0.886, 2.272, and 2.618.

Harmonic relationship Focus

Harmonic price zones occur when there is a convergence of harmonic price ratios/numbers or calculations that occur in a specific area. The trade reversals in these zones have a very high probability and can be traded with confidence. I have used harmonic price zones in the book to focus on certain trade examples.

Major Swing High or Swing Lows

Markets stage major support and resistance zones at key turning points called ‘swing highs’ and ‘swing lows.’ A major phase of support is established through a series of lower lows followed by higher highs forming a swing low and a resistance phase is established by higher highs followed by lower lows forming a swing-high. In this book, I focus on taking advantage of these support and resistance zones for either a primary or secondary target. I have used major swing highs or swing lows as protective profit targets. Most traders like to protect their profits at major swing high or swing low as the trade has a high probability of turning or pausing at these levels.

Chapter 1: Chart Types

1.1. Basic Charts

Basic Charts

Technical Analysis is the study of price and trend changes in Commodities, Stocks, Futures and various other market instruments. The price changes are primarily evaluated by various indicators, oscillators or trading systems to give a trader an edge in trading. Technical analysis is not a perfect science by any means, but it does have certain characteristics, patterns or indications which may be repetitive or may be intuitive and tend to possess Zen-like predictability power. Technicians plot these prices and price changes on a chart and apply various indicators and studies to figure out potential supply and demand areas, trade setups, targets and stops to win.

Technicians have developed various methods of representing market data on charts. The most extensively used charts are bar charts, line charts, candlestick charts and point & figure charts. There are many other variations like Kagi, Renko and Range bar charts. In this section, I will attempt to address the basic charts and their usage.

The most basic charts in technical analysis follow simple Cartesian structure (X&Y axes) to draw in 2-Dimensional space. On the X-axis (Horizontal), the time is plotted and on the Y-axis (vertical) the corresponding price is plotted. Any indicators derived from the time and price values, are either overlaid on the chart itself or plotted in secondary-graphs below and above the main price/time chart. Some traders plot volume on the X-axis as a representation of market activity.

Charts are plotted using various scales such as arithmetic or log /semi log charts. Arithmetic charts have the same distance between the prices where as log or semi log charts have a variable distance to represent the proportionate price movements.

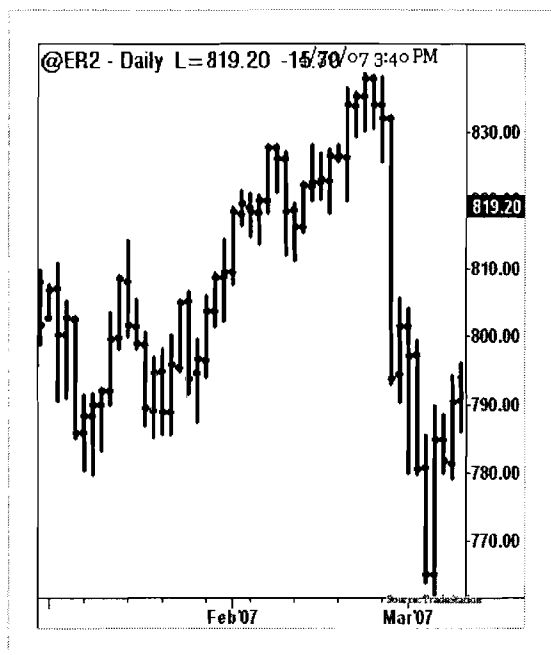
There are many facets in technical or chart analysis to understand and master. Price, Volume, and Time are the three most basic components of the market. Many successful traders only study price action to make money. Many other traders use complex mathematical theories and faster computer technologies to analyze and participate in the market action. Nevertheless, regardless of any trading theory or complex mathematical algorithm, the success in the markets lies with individual who can clearly understand the price-action and make the decisions to pull the trigger at the right time with excellent discipline. These individuals possess a higher understanding of market theories, market psychology and dynamics and money management methods and have mastered their execution skills.

Charts, patterns, indicators and software are only basic market tools. Successful traders view them just as tools and understand the usage. They build a theory and trade with a solid money management plan.

Basic Chart Types

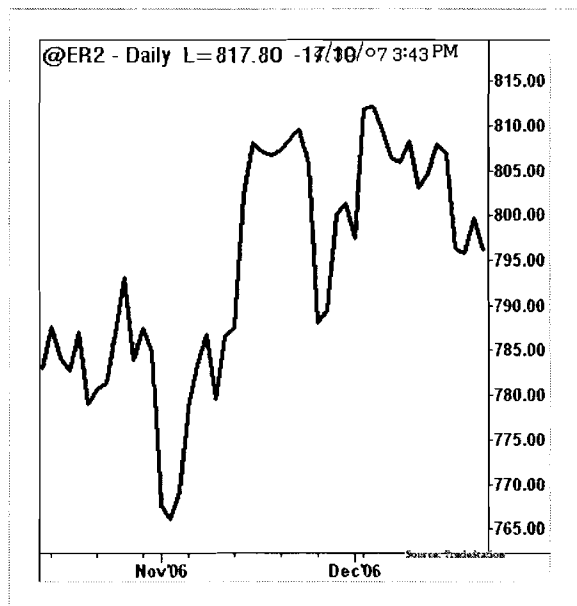
Bar Charts

Bar chart illustration is simple. Within each time-frame, a single vertical bar is plotted representing the price range within that time-frame. Each bar may have a left tick showing an Opening Price and a right tick showing a Closing Price. The top of the bar is the highest price reached within that time-frame and the low of the bar is the lowest price reached within that time-frame. The time-frame can be 1 tick, 1 minute, 1 day, 3-days, 1 week, 1 month or 3-months, or any finite numbers to represent time. The relationship between the Opening Price (left tick) and the Closing Price (right tick) represent the investor's sentiment and the trader's psychology within the trading session.



Line Charts

Line charts are based on "closing prices" only charts and have a cleaner look. They do not have open, high and low values plotted and potentially eliminate the noise and truly represent the value of the current price and true investor sentiment. Line charts can be plotted for highs, lows and open or pivot prices, but are only plotted for a single value object series in the entire chart. Line charts are usually plotted when there is a comparison between two market instruments such as spread charts, comparison charts and relative strength charts. Also, line charts are useful for small illustrations and trend displays. Most indicators are plotted using line charts, namely RSI and Stochastics.

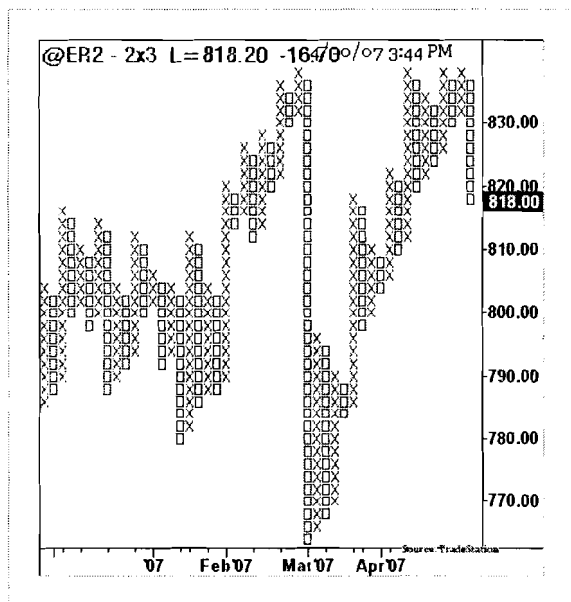


Basic Chart Types



Candlestick Charts

Candlestick charts were devised by Japanese rice traders in the 1600s and are discussed in detail in the next chapter. Candlestick charts are built on the open to close price relationships. The real-body is represented by the range between open to close and the color of the candle is black if the price closed below the opening price and white if the price closed above the opening price. The “wicks” on the both ends of candlestick represent the trading sentiment before settlement. Candlesticks have various patterns and truly represent supply and demand. Traders use candlestick charts with other market indicators such as moving averages, trend lines and RSI etc., to find the better opportunities than western charting methods such as bar and line charts.



Point and Figure Charts

The Point & Figure (P&F) charting technique is one of the oldest methods where the chart represents a true change (a box size) in price. In P&F charts, the time passage on the X-axis is ignored as the chart records the price change. P&F Charts have Xs and Os representing ascent and descent of a fixed box size price change. Each box size is preset and the price is represented only if the price trades above or below the previous box by that amount. The P&F method visually displays clear support and resistance levels. Trades are initiated from these levels on breakouts and breakdowns. Charts do not move in congestion range, hence choppy trading can be avoided. Box sizes can be a function of price range, average true range or a fixed size based on the closing prices.

1.2. Candlestick Charts

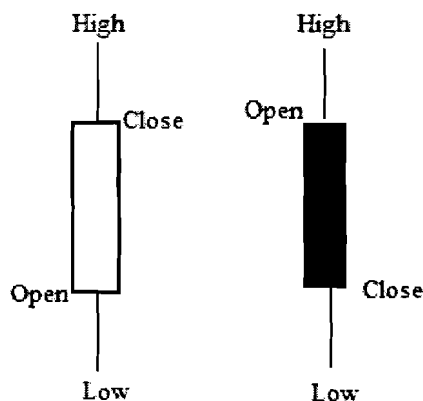
Candlestick Charts

Japanese rice traders invented Candlestick charting methods in the 1600's. "Candlesticks" show a visual representation of traders' emotions; whereas "bar" charts or "western" charts emphasize a focused approach on closing prices. Candlestick charts have a real-body (Open to Close) and shadows (Upper, Lower) showing intra-bar price relations between the key price values. In Candlestick charts, if a price closes higher than the open price then the Candlestick would be plotted Green suggesting bullish, and if the price closes lower than the open, the Candlestick would be Red, suggesting a bearish condition. The market sentiment is measured by the "real-body" length and its color. The bigger the real-body the bigger the sentiment and the smaller the real-body the smaller the sentiment which conveys indecision.

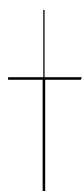
Candlestick charts offer a unique advantage over bar charts or line charts since they offer an excellent visual representation of the relationships with prior Candlestick bars. This indicates supply and demand along with the support and resistance levels, and possible trade decision opportunities for trend continuation or reversals.

Candlestick charts offer a simple way to show market movements and present outstanding trading opportunities. There are about 30-40 Candlestick patterns, continuation and reversals, which are helpful in trading. However, these patterns need to be clearly understood and mastered for successful trading purposes. The theory behind Candlestick charting method is not infallible. All patterns have clear confirmation theories and trading rules. The charts demand a full understanding of knowledge of pattern formations for successful trading.

There are many books written on Candlestick patterns and theories. Below are a few examples of trading Candlestick patterns.



Candlestick Patterns



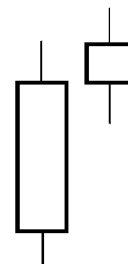
Doji



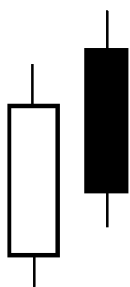
Inverted Hammer



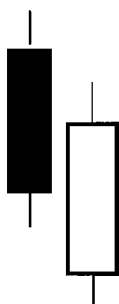
Hanging Man



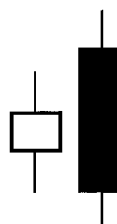
Spinning Top



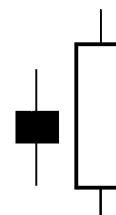
Dark Cloud



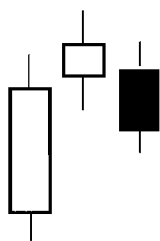
Piercing Lines



Bearish Engulfing



Bullish Engulfing



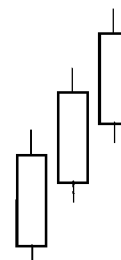
Evening Star



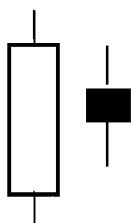
Morning Star



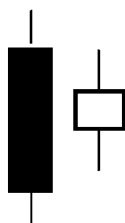
Three Black Crows



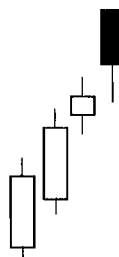
Three White Soldiers



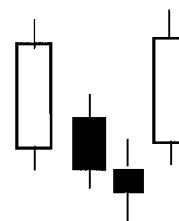
Bearish Harami



Bullish Harami

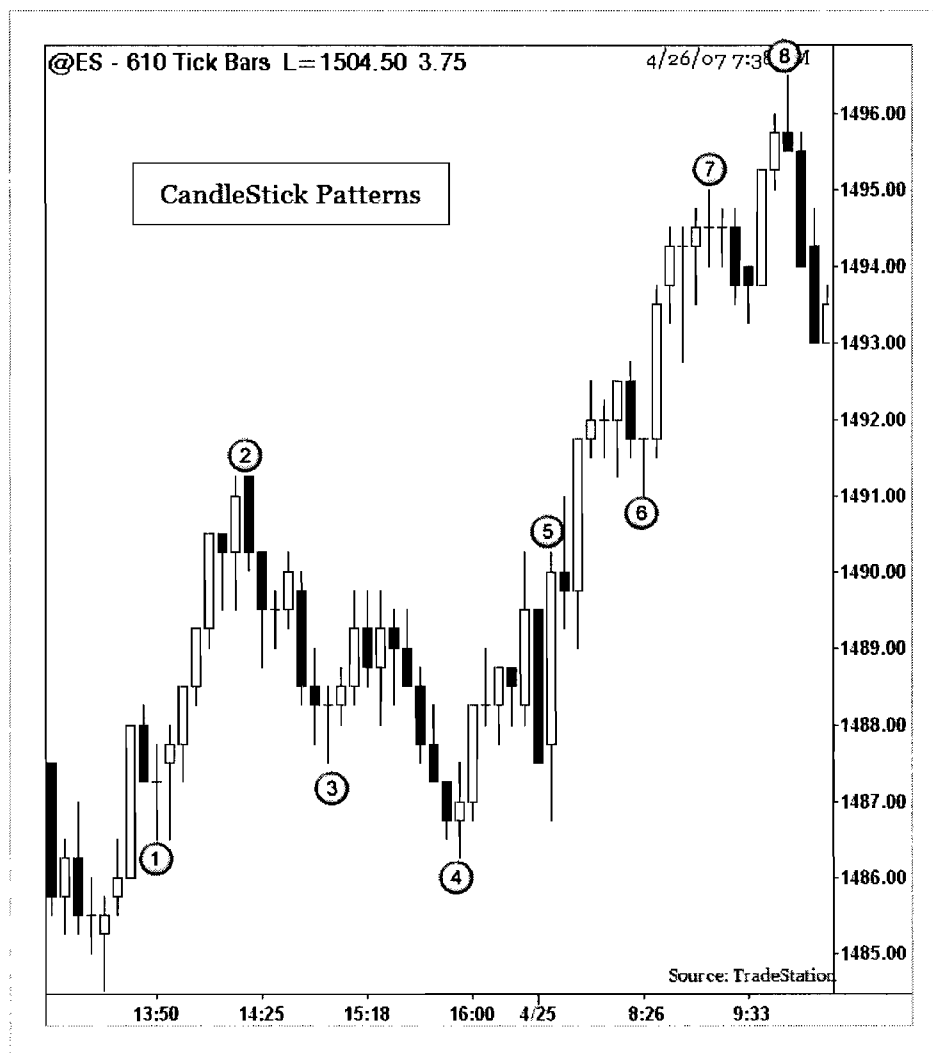


Belt Hold



Ladder Bottom

Trading Candlestick Charts



Trading Candlestick Patterns

The chart above illustrates various Candlestick patterns from the S&P 500 Futures 610 tick chart. Various Candlestick patterns have been marked in the chart above and explained as follows: 1). A Doji pattern to suggest indecision in the prior direction. 2). A Dark Cloud to signal a potential end of trend. (Also see Hammer pattern prior to the Dark Cloud). 3). Another Doji to signal indecision. 4). A Piercing line followed by a strong trend reversal bar. 5). A Bullish Engulfing pattern to confirm a strong trend ahead. 6). A Gravestone Doji. 7). A Doji bar to suggest an imminent trend reversal in the next few bars. 8). An inverted Hammer at the top to signal the end of uptrend and strong downtrend to follow.

1.3. Three Line Price Break Charts (3LPB)

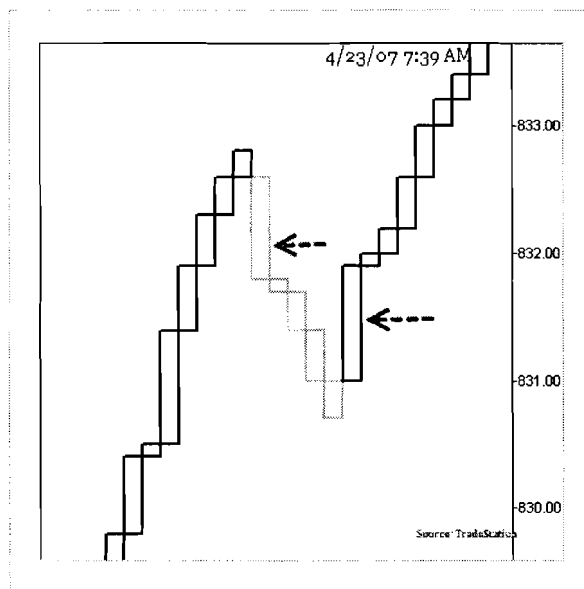
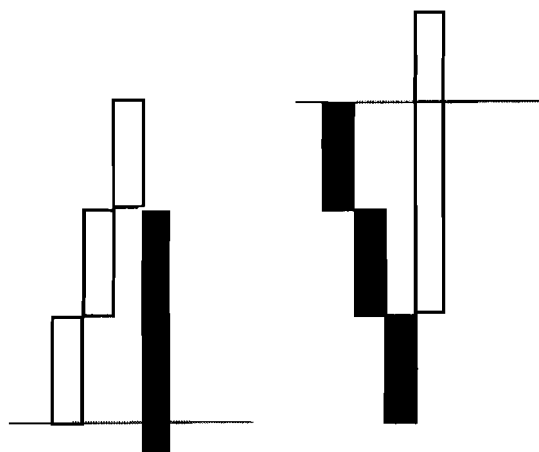
3LPB Charts

Three Line Price Break (3LPB) charts define the underlying trend and are considered as an adjunct to the candlestick charts. 3LPB charts display a series of vertical boxes or candles that are based on price changes. The 3LPB method entirely dispenses with the recording of the volume sales and time data on the X-Axis. The other major charting techniques like Point & Figure, Kagi and Renko charts also ignore the passage of time and volume.

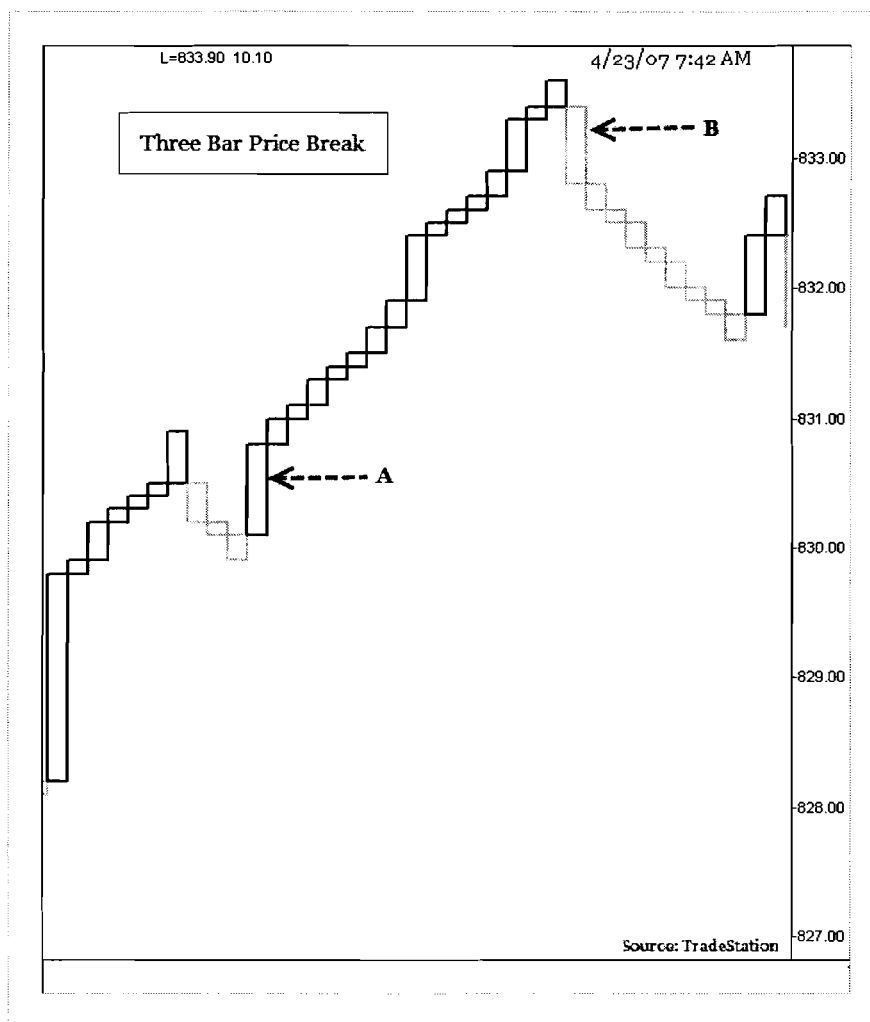
The 3LPB are always constructed based on closing prices. A basic understanding of 3LPB is when there are three white successive candles, the major trend is up, and when there are three successive black candles, the major trend is down. The major reversal signals (based on the 3LPB technique) are given when the shift lines, white to black or black to white are formed. After forming a confirmed trend of 3 white candles or 3 black candles, the reversal is only triggered if the current price is traded below the lowest of all prior three candles in case of a bullish trend reversal, or if the current price is traded above the highest of all prior three candles in case of the bearish reversal.

Trading with 3LPB

The 3LPB charts below show major trend changes. A trend change confirmation is made when a reversal bar is formed. However, a trend confirmation bar could be late and a significant move to the upside or downside may have already happened. A solution for this problem could be an intra-day trading signal for confirmation of the trend. The 3LPB charts also use other indicators and pattern formations to indicate price trends. The best trade signals are generated when the market reverses near the key support and resistance levels.



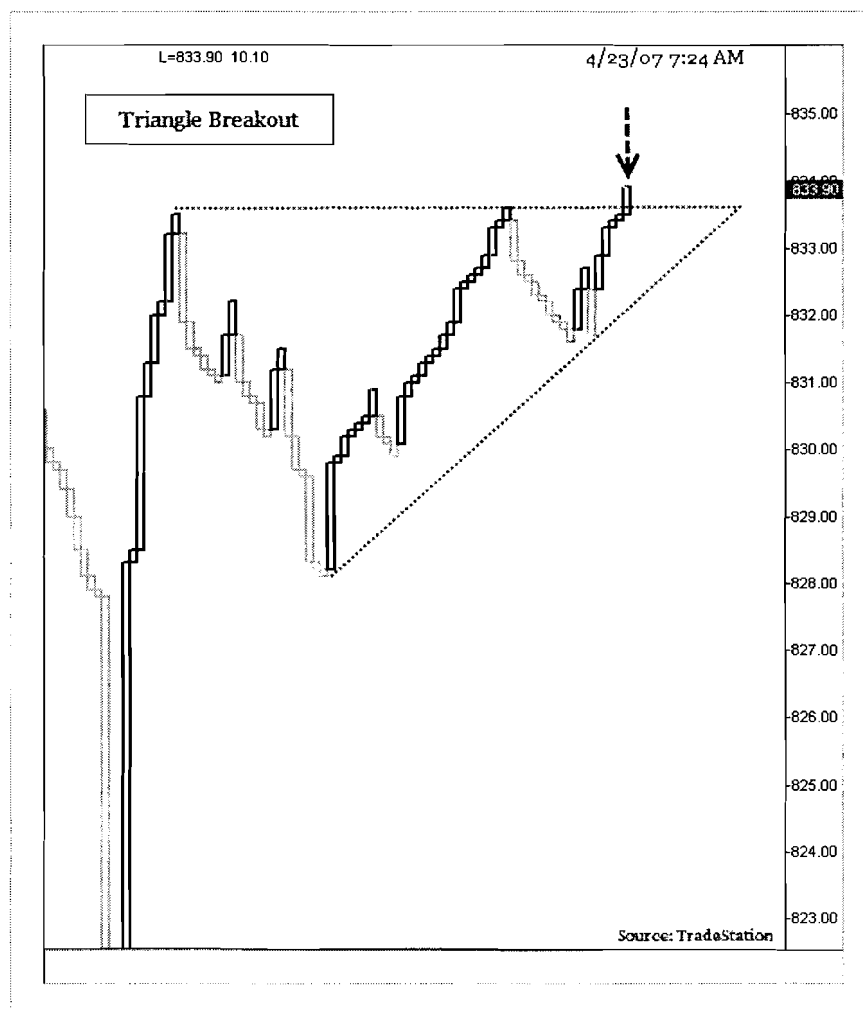
Trading 3LPB Pattern



Trading 3LPB Charts

The chart above shows an example of a 3LPB pattern from the Russell 2000 E-Mini one-minute chart. Candle A shows a Three Line Price Break reversal to the upside from the previous trend as the price closed above the high of the three candles. A "long" entry is placed at 831 level. A stop order is placed at the low of this breakout candle at 830. Targets are either taken with a reversal candle or at a pre-set target limit. One of the best techniques to place a target is to take the length of the reversal bar and use 2 times the length of the bar as the target. At candle B, a 3-Bar reversal candle is formed to enter a "short" position below the low of candle B. A "stop" order is placed above the high of Candle B. A target of twice the length of the candle is placed below the entry level.

Trading 3LPB Pattern



Trading 3LPB with Triangle Formation

The chart above shows a Three Line Price Break chart from the Russell Emini one-minute chart. The chart shows an ascending triangle pattern formation as the upper trend line was tested three times. The price closing outside the upper trend line signals a potential long trade.

1. Enter a "long" trade above the high of the breakout bar.
2. Place a "stop" order below the low of the breakout candle.
3. In an "Ascending triangle" pattern trade setup, the depth of the triangle is added to the breakout level for a target.

Chapter 2: Bar Groups

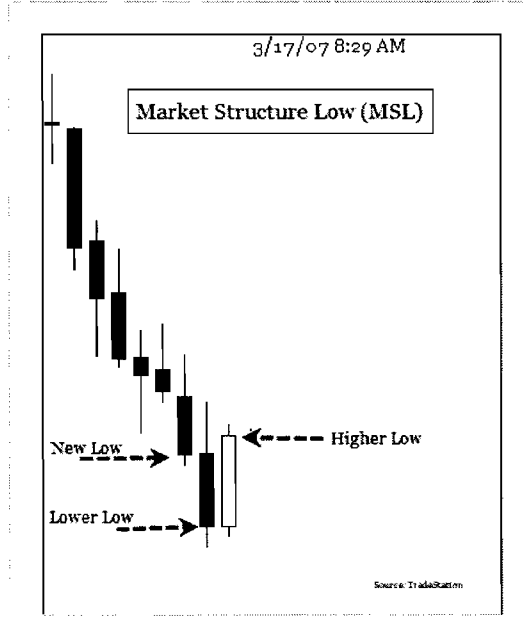
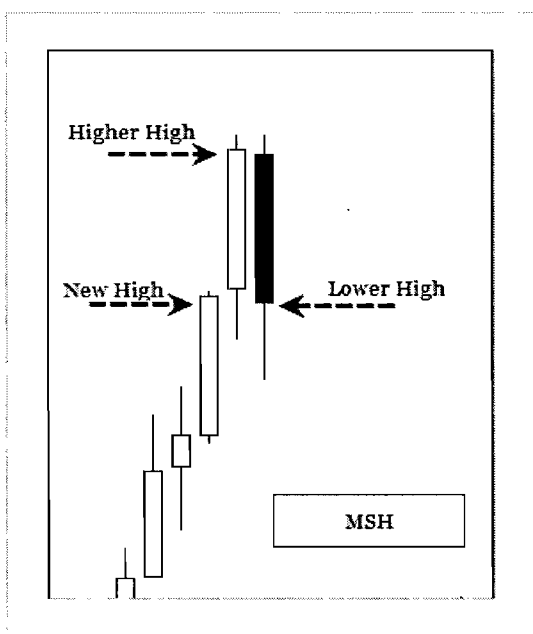
2.1. Market Structures

Market Structures

I first read about the Market Structures concept from Dr. Ron Lockhart. Then I read more detailed Market Structures work from Michael Jardine's book, *Fibonacci Trading*. The Market Structure concept is simple, yet a very powerful structuring concept. Markets have structures and traders miss most cues as they are constantly hunting some pattern, momentum, oscillator or some Zen-type signals. Market Structure formation is a price phenomenon occurring at major turns. Prices start with a wave (lets say Up) and it ends at some point and then there is a down-wave, and it ends some time and then begins another up-wave. These wave formations start with the Market Structures. Market Structure Low (MSL) and Market Structure High (MSH) formation is continuous and is repetitive at every wave begin and at every wave end.

Market Structures form in all markets, in all time-frames and in all instruments. They fail and re-fail, form and re-form. Market Structure is a concept pattern. It needs other indicators, support/resistance levels, triggers to confirm the theory and its works.

Market Structures are well explained with Candlestick charts. Market Structure Low (MSL) is explained with three candle pattern. A new low, lower low, higher low of **CLOSE**. Please see the word in **BOLD**, MSL is based on close values *not* lows or highs. A MSL based long is triggered when prices close above the highest close value. A stop is being placed below the low of the MSL to protect the trade.



Market Structure High (MSH)

A Market Structure High (MSH) is formed when markets are making major tops. They form near key resistance areas or key moving average levels, which is very significant. There are many theories of using MSHs in trading. One theory of MSH, is to find a critical resistance area and trade a short position. The other way is to find a series of past MSHs and build a trend line for resistance. The third way is to use MSH in a fractal wave form and build a larger wave structure using the distances between them.

The Market Structure High (MSH) definition is when markets make a new high followed by higher high followed by lower high. This pattern is shown with a three candle pattern. But many cases it may not be. It can form in 5-6 candles in a time-frame, but the internal candles should be mostly inside-bars suggesting indecision.

Trade:

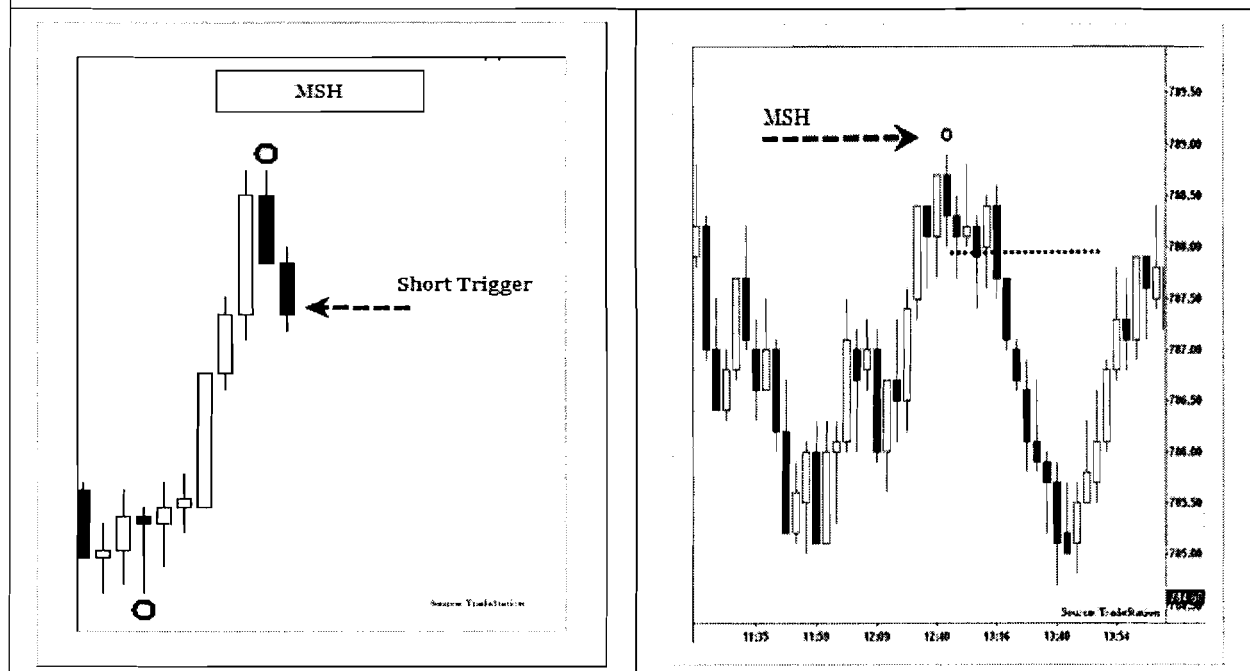
A MSH short trigger (after 3-bar MSH formation) is signaled when price closes below the low of the third candle.

Stop:

Place a "stop" order above the MSH pattern's high.

Target:

There are two ways to target MSH shorts. One is to trade until another MSL forms near a key significant level. The second method is to trade with a trailing stop using the previous bar's high as the stop.



Trading MSH



Trading Market Structure High (MSH)

The example above illustrates a Market Structure High (MSH) formation from the ER2 610 tick chart. On March 14, 2007, at around 9:35am, ER2 formed a Market Structure High to signal a potential top at 780 level. A short trading opportunity is presented when prices closed below the low of the third candle.

1. Enter a "short" trade below the low of the third candle at 778.
2. Place a "stop" order above the MSH high at 780.2.
3. Target is dynamically changed and traded until the prices form either another MSL, or if the price closes above the previous bar's high after an initial profit.

Trading MSL

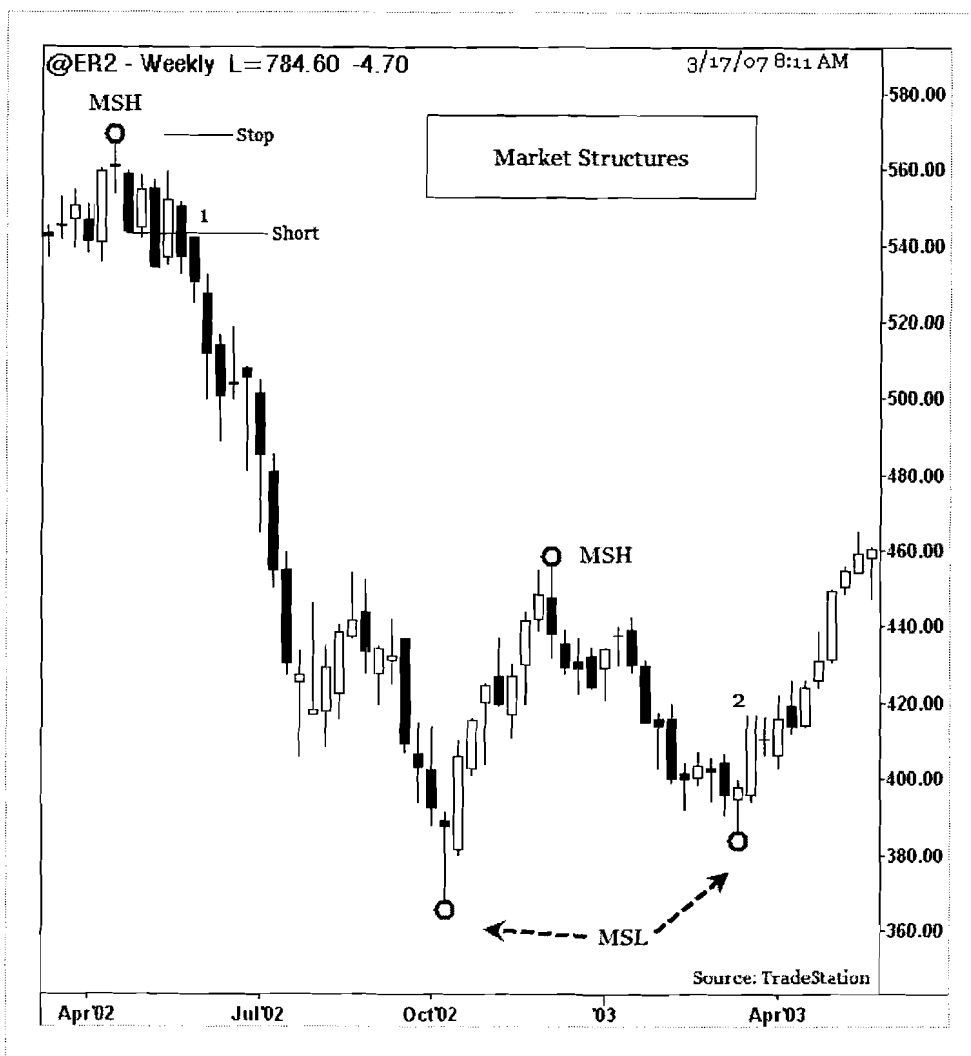


Trading Market Structure Low (MSL)

The example above shows a MSL formation from the Dow EMini futures (YM). In October, 2005, YM made a MSL formation after a long downtrend around 10700 level. This MSL is confirmed when the price closed above the high of the third candle.

1. Enter a "long" trade above the high of MSL at 10850.
2. Place a "stop" order one-tick below the low of the MSL at 10650.
3. Target a reversal MSH formation or when price closes below the low of previous bar.

Trading Market Structures



Trading Market Structures

The example above illustrates a Market Structure trading formation on Russell Emini futures (ER2) weekly chart. In April 2002, ER2 formed a Market Structure high at 560 level. A short trading signal was triggered the following week at 540 level. A stop is placed above the high of MSH at 565. A profit level for MSH trading is set either at a previous swing low or MSL level, or until another MSL is formed to reverse its trend direction. A similar trading opportunity for MSH is presented again in December 2002. A MSL trading opportunity is shown in April 2003 around 420 levels. A stop loss is set at MSL low (at 390).

2.2. Three Bar Groups

Three Bar Groups

Chart formations in Technical analysis require a group of bars to derive a pattern. Sometimes a single bar or two bars can show great patterns, but as a group, 3-bar series groups provide reliable patterns or confirmations for other major developing patterns. These groups of bars are called “key reversal” bars. This 3-bar group may also consist of well known two-bar reversals or a single bar patterns within inside this group. Most of these 3-bar groups are part of a “fractal” formations or part of “market structures” where a prevailing trend showing signs of pausing or reversal of current trends. Bars with exhaustion price-action, “narrow range (with inside-days)” or “spike with ledges” are some of the 3-bar group pattern examples. 3-bar group pattern formations near key support and resistance levels or near key moving averages (50 EMA, 200 SMA) offer great potential trade setups.

Within the 3-Bar Groups, intra-bar relations like close and open values relative to other bar close and open values and how they are formed could give signals of continuation or reversal of trends. Gaps within the 3-bar patterns also have significance.

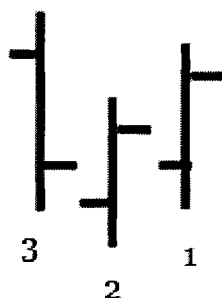
Three Bar Pattern groups as the name suggests, will have three continuous bars. It can be in any time-frame or in any market instrument. A 3-bar group pattern is defined using the three bar’s inter-bar Open, High, Low, Close relationships with each other. In my view, Three Bar patterns are relatively short trade setups and should be traded using other indicators. They are more effective as reversals near the end of prolonged trends than in the middle of the trends. When markets making new highs and showing a series of signs of pausing or reversals, 3-bar patterns are more reliable than 3-bars formed in the middle of the trend. Two out of three 3-Bar Groups may be successful but the concept also applies to bigger structures with three continuous major “swing highs” and three continuous major “swing lows”. When trading three-bar groups, look for the third bars’ range. When the range of the third bar is greater than prior two bars, it tends to produce more reliable results.

One of the 3-Bar Groups (Market Structures) is discussed in detail in this book and here I present few of my favorite patterns that I trade.

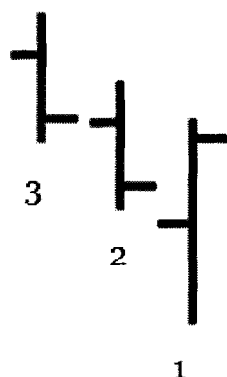
Trading 3-Bar Groups

All the 3-Bar group patterns listed below have trade-setups. Most of these patterns are short-term based and targets are usually at a major “swing high” or major “swing low” based on the pattern setup. Stop orders should be placed to protect the trade within the 3-bar groups. Three bar group patterns fail when significant support or resistance is traded against the trade setup. When trading an upside 3-bar group, place a stop order below the lowest low of the three bars. When trading a downside 3-bar group, place a stop order above the highest high of the three bars.

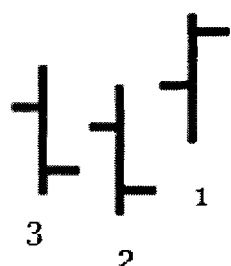
Trading Three Bar Groups



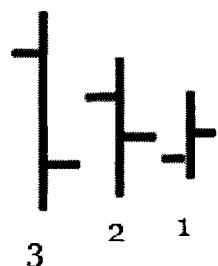
$H[3] > H[2]$
 $C[2] < C[1]$
 $L[1] > L[2]$
 $H[1] > H[2]$
 Trade above $H[3]$



$H[3] > H[2]$
 $C[2] < C[1]$
 $L[2] < L[3]$
 $O[1] < C[2]$
 $C[1] > C[2]$
 $L[1] < L[2]$
 Trade above $H[2]$

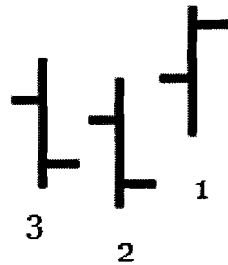


$H[3] > H[2]$
 $C[2] < C[3]$
 $O[1] > C[2]$
 $H[1] > H[3]$
 Trade above $H[1]$

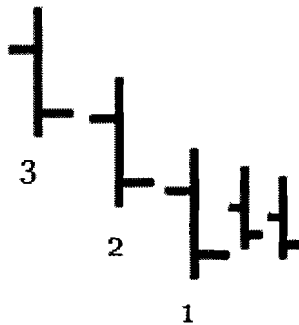


$H[3] > H[2]$
 $H[2] > H[1]$
 $L[3] < L[2]$
 $L[2] < L[1]$
 Trade above $H[3]$
 or Trade below $L[3]$

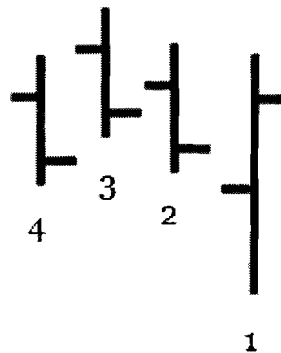
Trading Three Bar Groups



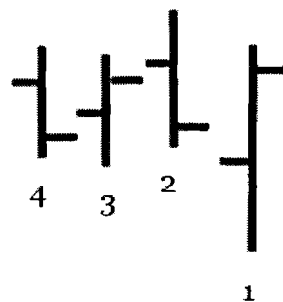
$H[3] > H[2]$
 $C[2] < C[3]$
 $O[1] > C[2]$
 $H[1] > H[3]$
 Trade above $H[1]$



$L[2] < L[3]$
 $L[1] < L[2]$
 $H[3] > H[2]$
 $H[2] > H[1]$
 Series of Small bars
 Trade below $L[1]$



$H[4] < H[3]$
 $C[3] > C[2]$
 $C[2] < C[3]$
 $O[1] < C[2]$
 $C[1] > C[2]$
 Trade above $H[3]$



$H[3] < H[4]$
 $C[3] > C[4]$
 $H[2] > H[3]$
 $C[2] < C[3]$
 $L[1] < L[3]$
 $C[1] > C[3]$
 Trade above $H[2]$

2.3. Matching Highs/Lows

Matching Highs/Lows Pattern

Matching Highs and Matching Lows patterns occur at market bottoms and tops. Prices form key support and resistance areas at certain price ranges, and attempt to break these levels for at least 3 bars in a row. When the prices fail to break these levels, they form “matching lows” and matching high patterns for a minimum of 3 bars, and these “matching highs” or “matching lows” may signal a potential trend reversals. These patterns are more effective in daily and weekly charts. Intra-day tick charts with higher tick counts or time-frames are used when finding reliable matching high/low patterns.

Trade:

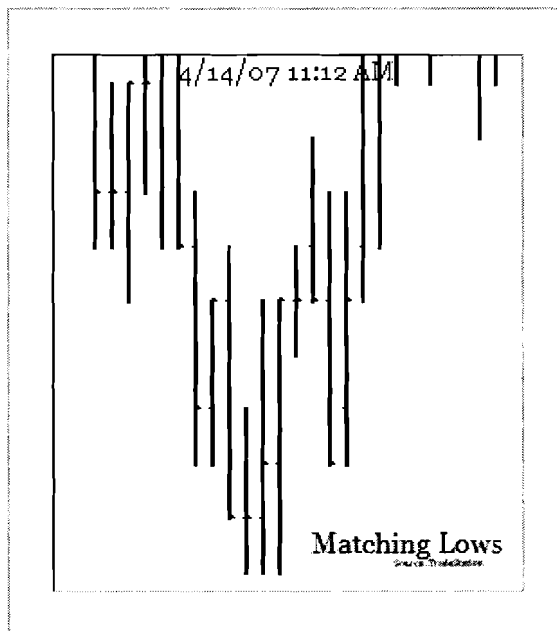
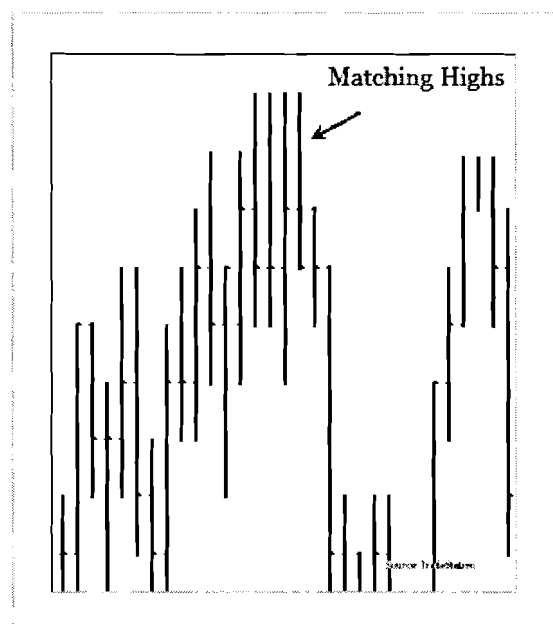
Trades are taken in the opposite direction prior to the matching highs/lows pattern. After a “matching lows” pattern, a “long” trade may be entered above the high of the breakout bar. For a matching highs pattern, a “short” trade is entered below the low of the breakdown bar.

Stop:

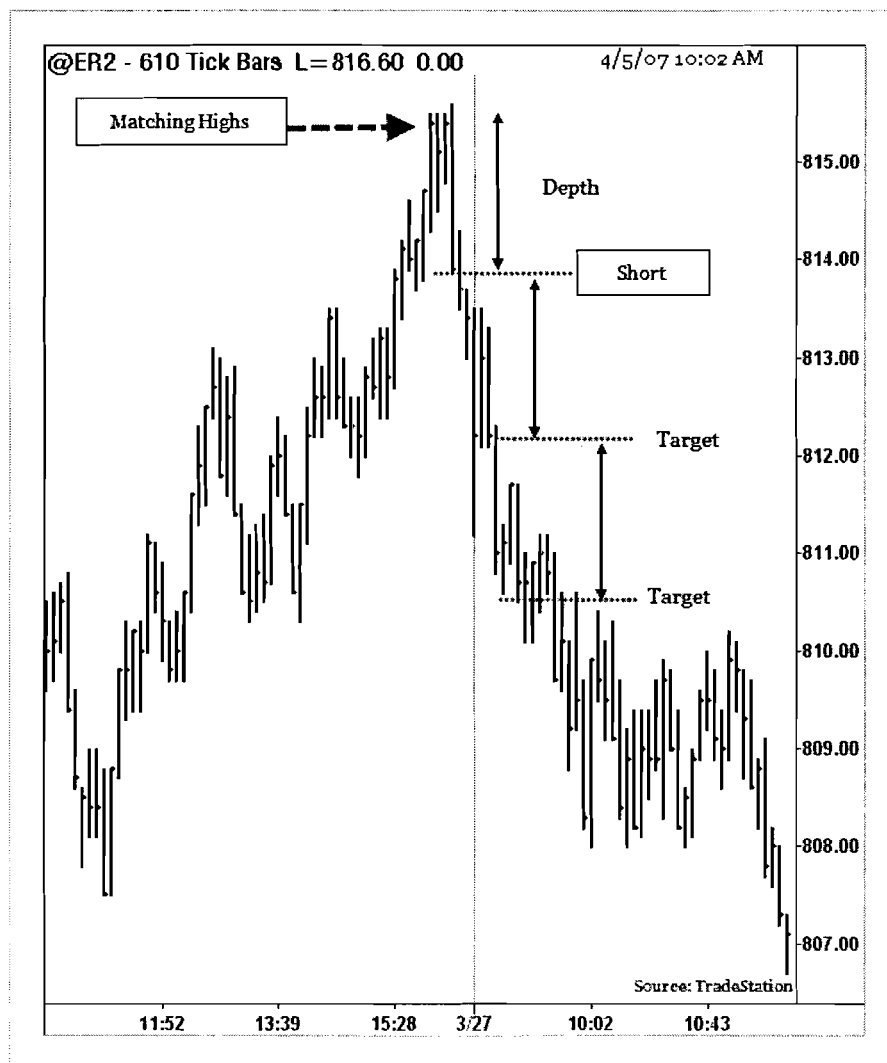
Place a “stop” order below the low of the pattern for a “matching lows” pattern. For a matching highs pattern, place a “stop” order above the high of the pattern.

Target:

Targets are placed from the trade entry in multiples of length of the breakout bar. For a long entry, place the first target at the length of the breakout bar above the entry, and the second target is set at twice the length of the breakout bar above the entry. For short entry, place the first target at length of the breakout bar below the entry level, and the second target is set at twice the length of breakout bar below the entry.



Trading Matching Highs Pattern

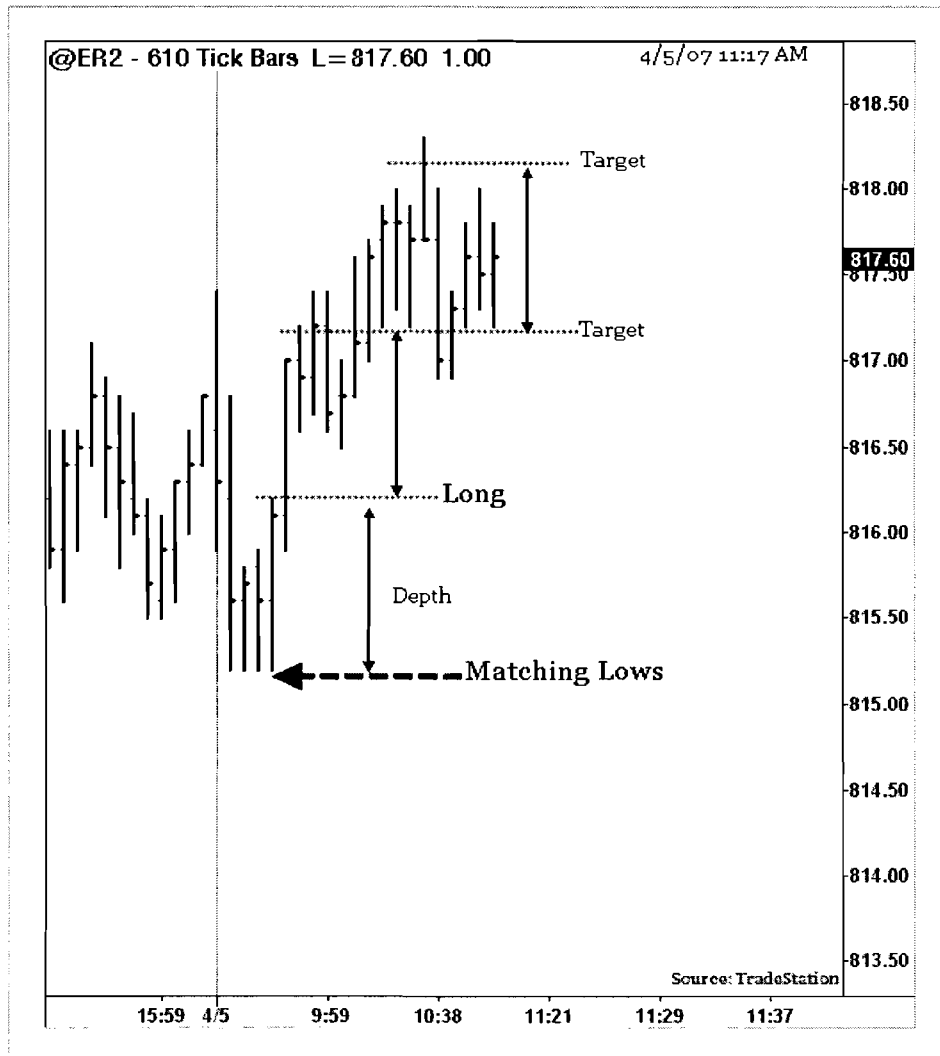


Trading Matching Highs Pattern

The example above illustrates a “Matching Highs” pattern from the Russell Emini futures (ER2) 610 tick chart. On March 26, 2007, ER2 rallied most of the day to form a top at the 815.5 area. Late in the afternoon, Russell formed a Matching Highs pattern for a series of bars to suggest a potential short trade.

1. Enter a “short” trade below the low of the breakdown bar at 813.6.
2. The length of the breakdown bar is measured at 1.6 points.
3. Set a profit target at 812 (length of the breakdown bar) and 810.4.
4. Place a “stop” order above the high of the pattern at 815.5.

Trading Matching Lows Pattern



Trading Matching Lows Pattern

The chart above illustrates an example of a Matching Lows pattern from the Russell Emini futures (ER2) 610 tick chart. On April 5, 2007, during the morning trading, ER2 formed a matching lows pattern at 815 area to suggest a potential long trade. Wait for a breakout bar to confirm higher highs before placing a trade.

1. Enter a "long" trade one-tick above the breakout bar's high at 816.3.
2. The length of the breakout bar is 0.8 points.
3. Set targets at 817.1 (length of the bar) and 818.
4. Place a "stop" order below the low of the matching low pattern at 815.1.

2.4. n-Bar Rallies/Declines

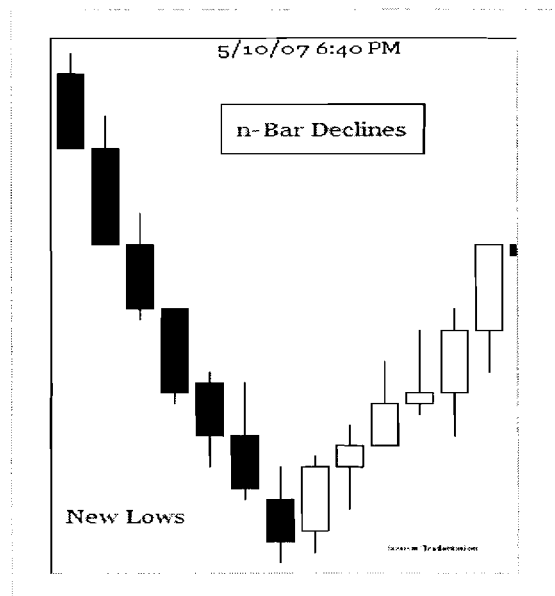
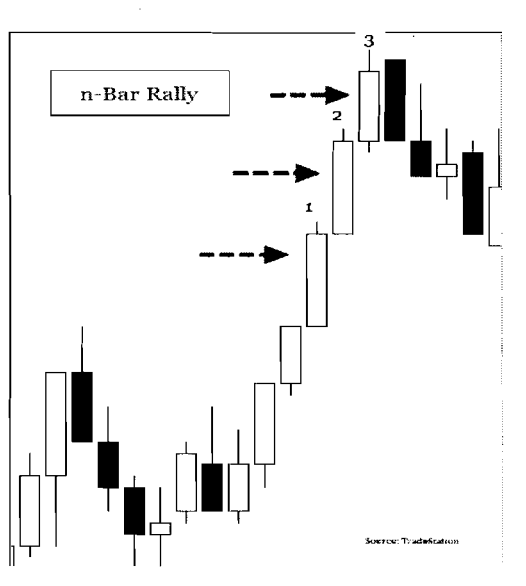
n-Bar Rallies/Declines

Stocks making new Highs and new Lows with high volume attract momentum players. Momentum players continue to push these stocks until the supply or demand diminishes. But as in any rally or decline, these trends stop and reverse. The prices become oversold and overbought creating a condition for exhaustion of the supply and demand. Hence, if these prices reverse (could be brief), an opportunity is presented. Traders who wait and take advantage of these counter-trades are daring and risky, but they do present a healthy risk/reward profit ratio. The popular trading strategy is to wait for the trend to stop and reverse. The n-Bar Rally/Decline setup presents a technique of trading these counter-trend setups.

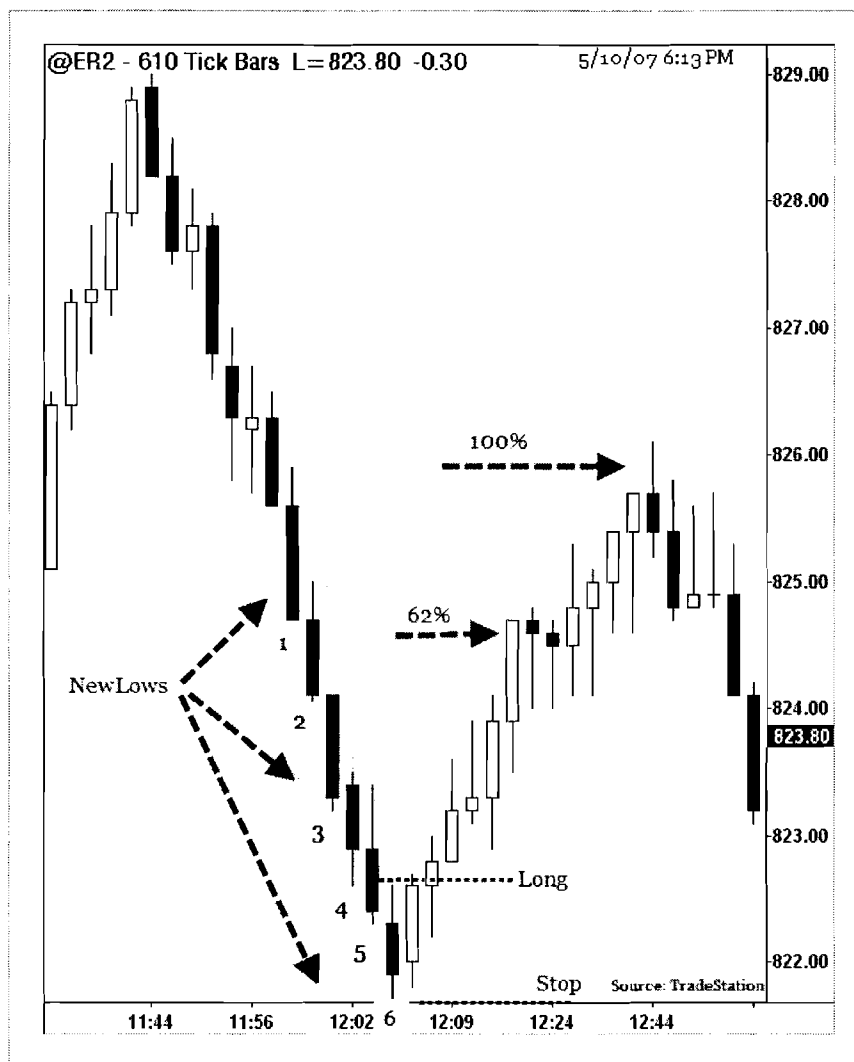
Trade: The n-Bar Rallies/Declines build up on based on new-highs and new-lows setups. Prices must make at least a 21-bar new high and new-low, and at least 3 new highs or 3 new lows successively-the higher the number, the better the pattern result. After an exhaustion move, enter a long one tick or 5 cents above the last falling bar's high, or enter a short one tick below the last rising bar's low.

Stop: Place a "stop" order one tick below the low of the falling bar in a long setup, or place a "stop" order one tick above the high of the rising bar in a short setup.

Target: Place a target 62% to 100% of the entire range of n-bar Rally/Declines for long or short setups. The other targets could be a major "swing high" or major "swing low" prior to the n-Bar setup.



Trading n-Bar Rallies/Declines

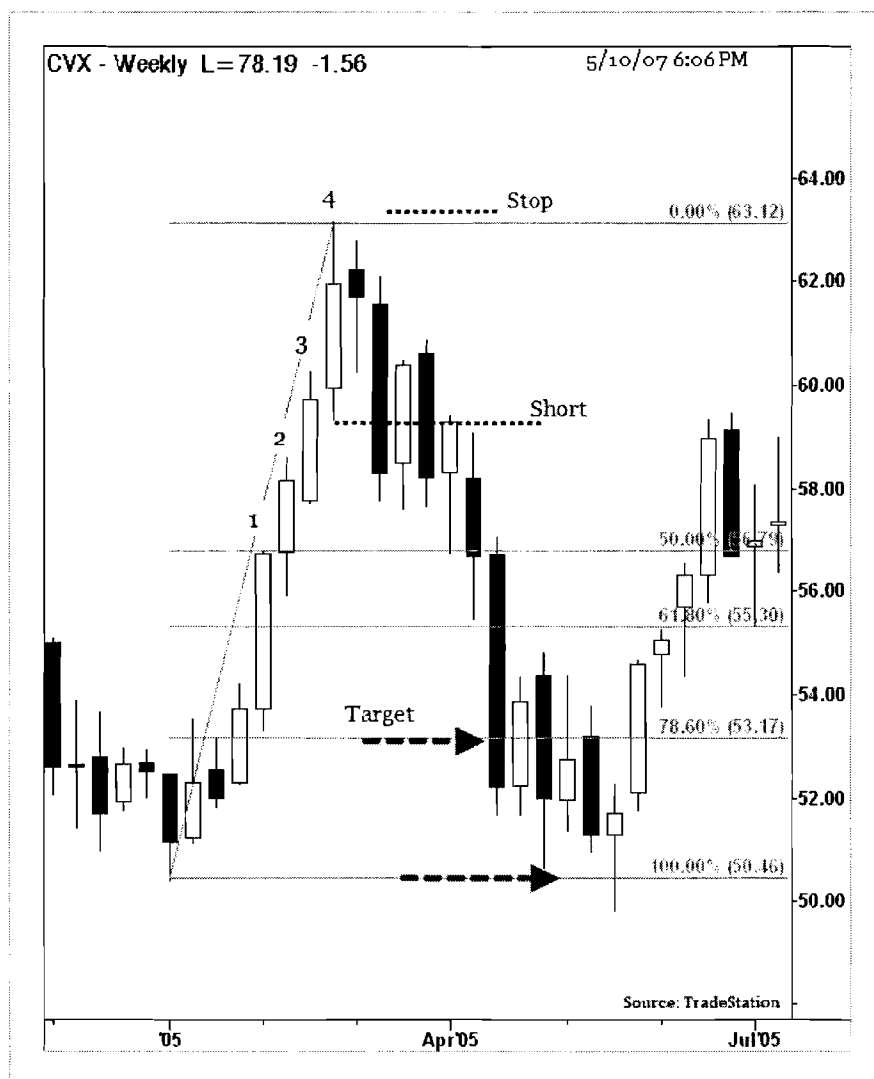


Trading n-Bar Declines

The example above illustrates an n-Bar Decline pattern from the Russell 2000 Emini chart. On May 10, 2007, ER2 sold off in the pre-afternoon session and started to make new lows around 12pm. Six new lows bars were traded before a reversal bar triggered a trade above the previous bar's high. This scenario presented a trade long in the downtrend.

1. Enter a "long" trade one tick above the previous bar's high at 822.7.
2. Place a "stop" order below the low of the n-Bar decline at 821.6.
3. Target between 62% and 100% of the n-Bar decline range from 824.8 to 826.

Trading n-Bar Rallies/Declines



Trading n-Bar Rally

The example above illustrates an n-Bar Rally setup from the Chevron weekly chart. From January to March of 2005, Chevron's stock rose from \$52 to \$63. Chevron made a series of higher-high prices to complete n-Bar High pattern. At the beginning of April, 2005, Chevron stopped making new highs and presented a potential short setup.

1. Enter a "short" trade below the low of the last higher-high bar at \$59.70.
2. Place a "stop" order above the high of the last n-Bar setup at \$63.25.
3. Place the target from 62% to 100% of range of the n-Bar setup from \$55 to \$53.

2.5. Seven-Day Narrow Range & Inside Bar & Opening Range Breakout (NR7ID with ORB)

NR7ID with ORB

Toby Crabel popularized a trading pattern called NR7ID with ORB in his book, *Day Trading with Short Term Price Patterns and Opening Range Breakout*.

7-Day Narrow Range (NR7): NR7 is defined as the daily range which is narrower than the prior 6-Day range.

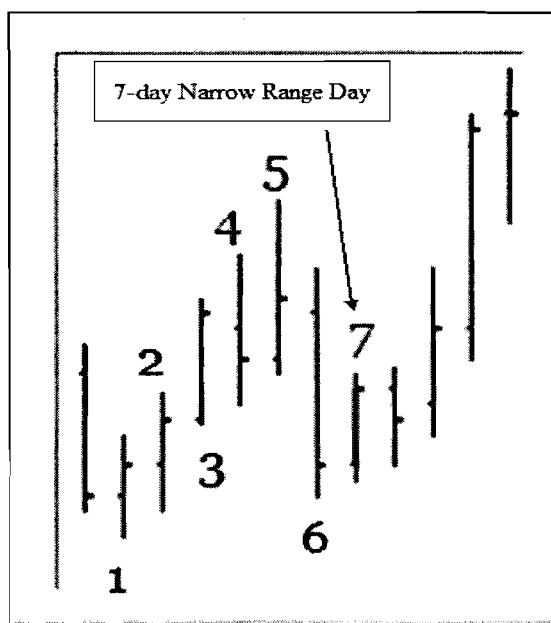
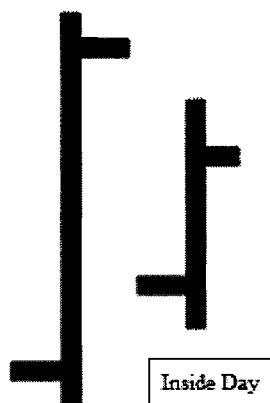
Inside Day (ID): Inside Day is defined as a trading day that has range which is completely encompassed by the previous days' range. The prior day's high is higher than the current days' high and the prior day's low is lower than current days low.

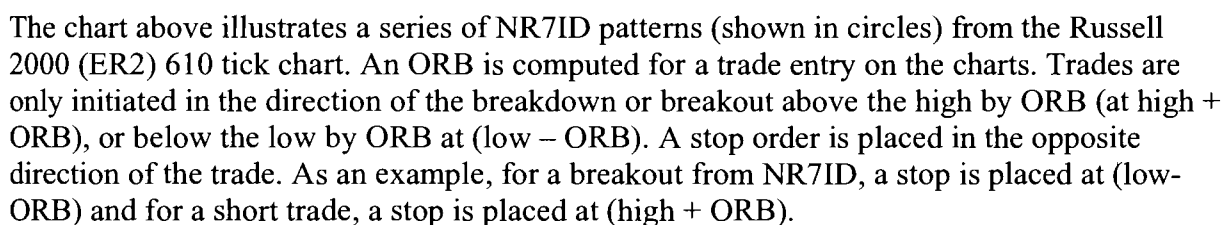
Opening Range Breakout (ORB): Is a trade executed when prices trade a predetermined amount above or below the opening range. The predetermined range is calculated as the 10-bar average distance between open to high and open to low.

Trade: Enter a trade after the NR7ID in the direction of the breakout. Enter a long trade on breakout at the ORB value above the high. For a short trade, enter a short trade on breakdown at the ORB value below the Low.

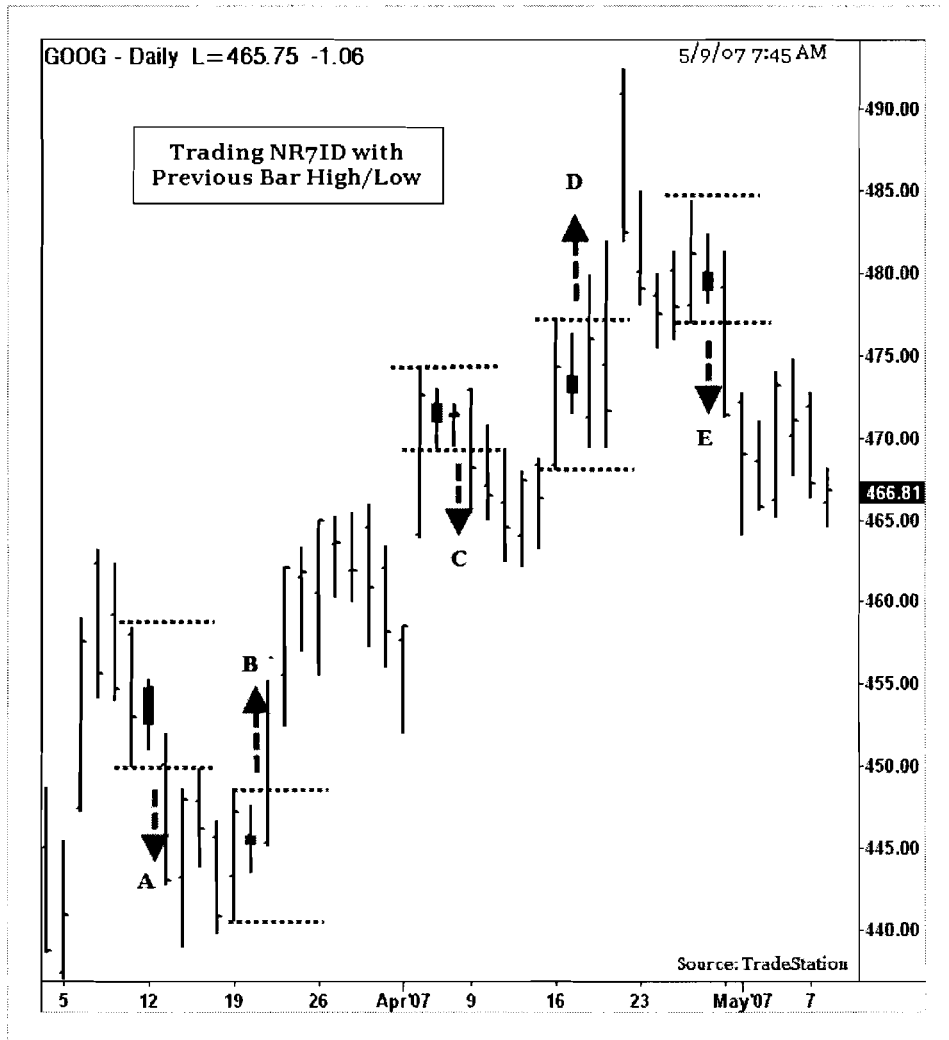
Stop: For breakout trades, place a stop order at (low-ORB). For breakdown trades, place a stop order at (high + ORB).

Target: "NR7ID with ORB" is primarily a trade entry technique and the targets are set at prior "swing highs" and prior "swing lows" or at key resistance and support areas.





Trading NR7ID



Day Trading NR7ID with Previous bar High/Lows

The chart above illustrates an example of a NR7ID pattern with previous bar highs/low trading from the GOOG daily chart. An efficient way to trade NR7ID bars is to use the previous bar high, or the previous bar low as the trigger points. Trade stops are placed on the other end of NR7ID against the trade. Usually these trades are very short-term and should be closed within 1-3 bars (days). In the chart above, on March 12th, GOOG formed a NR7ID pattern at A. A trade setup is to enter a long trade on the next day if the price trades 10 cents above the previous days high or enter a short if the price trades 10 cents below the previous days' low. Trades at B and D are long trades. Trades at C and E are short trades.

2.6. Seven Day Wide-Range and Outside Day Pattern (WR7OD)

WR7OD Pattern

A 7-day Wide Range bar is formed when a current bar has the widest range in the last 7 bars. An outside bar develops when the low of the current bar is lower than the previous bar and the current high is higher than the previous bar. Trades are only entered in the direction of the current trend.

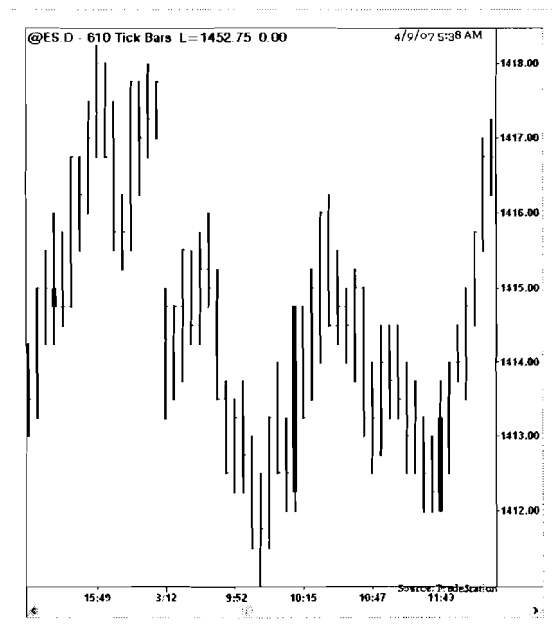
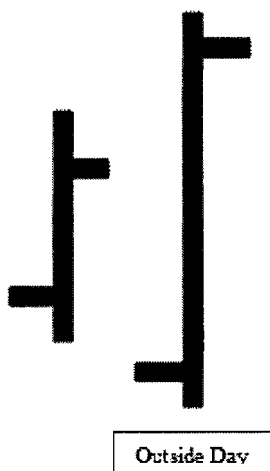
Wide Range bars forming at the start of a trend or at key reversal levels could signal strong bullish and bearish trends. If Wide-Range bars are forming out of a consolidation range, they signal price continuation in the direction of breakouts. And Wide-Range bars forming at the end of the rallies and sell-offs, the bars may be signaling exhaustion and potential trend reversals.

Trade: Wait for WR7OD to form in the current up or down trend. In an uptrend, trade only “long” one tick above the high of the WR7OD bar. In down trends, trade only “short” one tick below the low of the WR7OD bar.

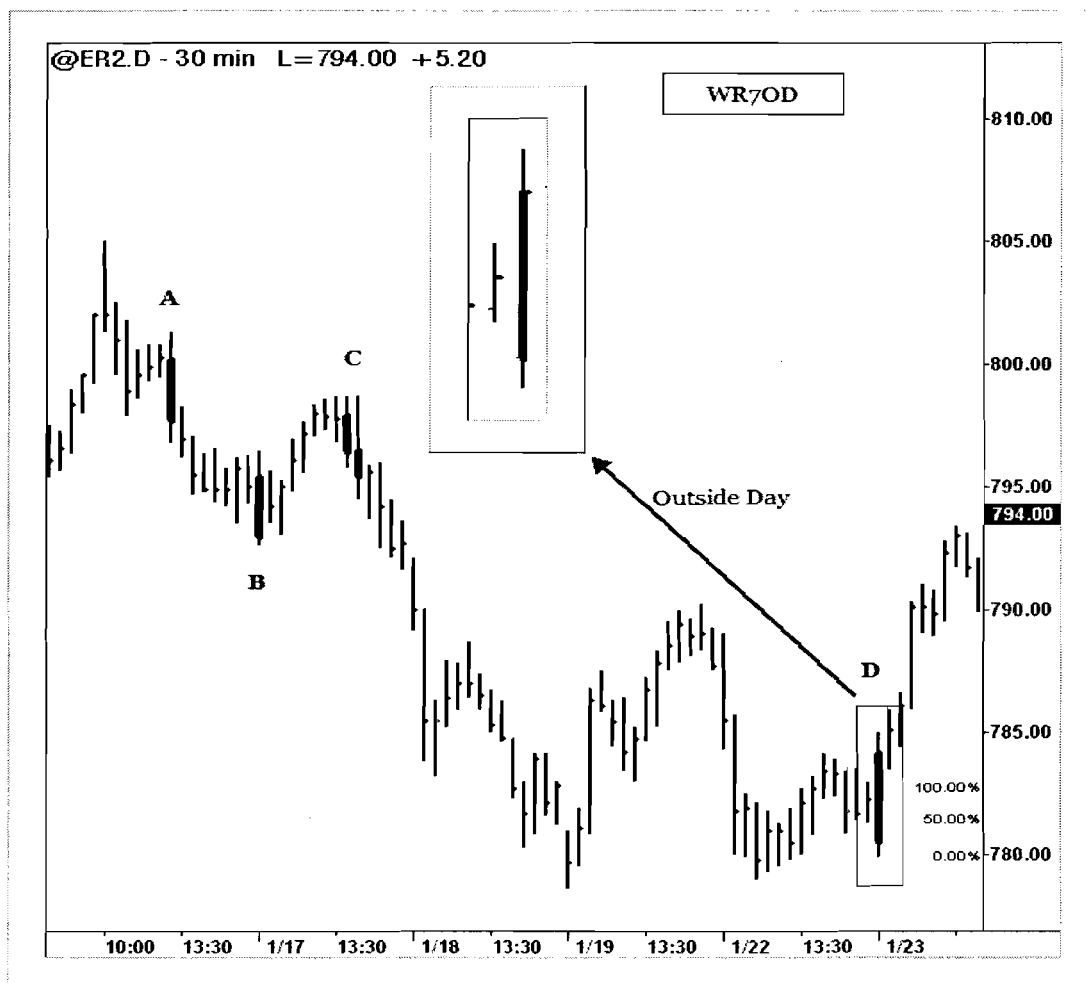
Stop: Place a “stop” order one-tick below the low of the WR7OD bar (for breakouts). Place a “stop” order one-tick above the high of the WR7OD bar (for breakdowns).

Target:

Set targets of 50% to 100% of the WR7OD range from the breakout or breakdowns levels.



Trading WR7OD Pattern



Trading WR7OD Pattern

The example above illustrates a WR7OD pattern from the Russell Emini (ER2) 30 minute chart. Please note the Wide Range bar inset. Wide Range bars are only traded in the breakout direction of the trends. A 7-day Wide Range bar signals trend shift. Long trades are initiated high above the Wide Range bar. In the example above, bars A, B and C are in down-trend and trades are only taken from downside. Bar D is formed in uptrend and trade is only taken in the upside direction.

1. Enter a "long" trade one-tick above the high of the WR7OD bar.
2. Place a "stop" order one-tick below the low of the WR7OD bar.
3. Place targets from 50% to 100% of the WR7OD range from the breakout level.

Trading WR7OD Pattern



Trading WR7OD Pattern

The example above shows a 7-day Wide-Range with outside bar formations from the S&P futures 610 tick chart. WR7OD bars signal the exhaustion and trend reversals. Trading WR7OD coupled with any confirmation indicators, or other market patterns, could be profitable. The first WR7OD bar was formed around lunch time in an uptrend. A “long” trade can be entered above the high of the WR7OD bar at (1). The second WR7OD bar is formed at around 1:30 pm in downtrend. A downside trade is triggered below the low of the Bar 2. The third WR7OD bar is a trend reversal bar and trade is triggered above the high of the WR7OD bar. The fourth WR7OD bar is a continuation bar similar to the third WR7OD bar.

Chapter 3: Pivots

3.1. Floor Pivots

Floor Pivots

Traders believe the market direction can be divulged by Floor Pivot levels. They buy or sell these pivot levels and exit at either the first or second resistance levels. They also have the option to sell the pivot level and cover at the first or second support levels. Floor pivot trading is an effective way to find support and resistance levels, and is widely used by many traders. Prices around pivot levels signal choppy or trend market modes. In bullish markets, when prices consolidate around the pivot levels, prices tend to pick the prior trend direction and trade higher. Pivot trading is very crucial and most traders utilize daily Pivots for trading. However, weekly and monthly Pivots are equally important. Some traders compute the mid points between the Pivot and support or resistance levels and plot them on their charts.

A confluence of pivots increases the chance of potential support/resistance compared to a single pivot level. Often there will be many confluences between daily and weekly pivots on a chart and these levels could be significant for traders.

It is almost rare to see price reaching beyond R3, R4 or S3 and S4 levels on regular daily trading. Also, it may be rare to see prices outside of weekly S2 and R2 levels. Some traders use weekly extreme levels and short weekly "R2" level and cover at the weekly Pivot or buy the weekly "S2" level and sell at weekly Pivot levels.

Pivots are very efficient for both day and swing trading. Pivot trading is quite profitable using these support/resistance levels in the direction of the trend along with good money management techniques.

Floor Pivot Calculations

Pivot Point (PP) = $(H+L+C)/3$

First resistance (R1) = $2*PP - L$

First Support (S1) = $2*PP - H$

Second Resistance (R2) = $PP + (R1-S1)$

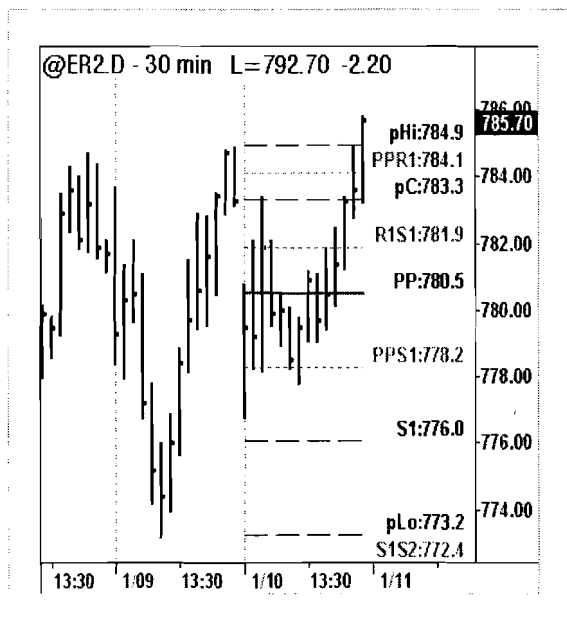
Second Support (S2) = $PP - (R1-S1)$

Third Resistance (R3) = $R1 + (H - L)$

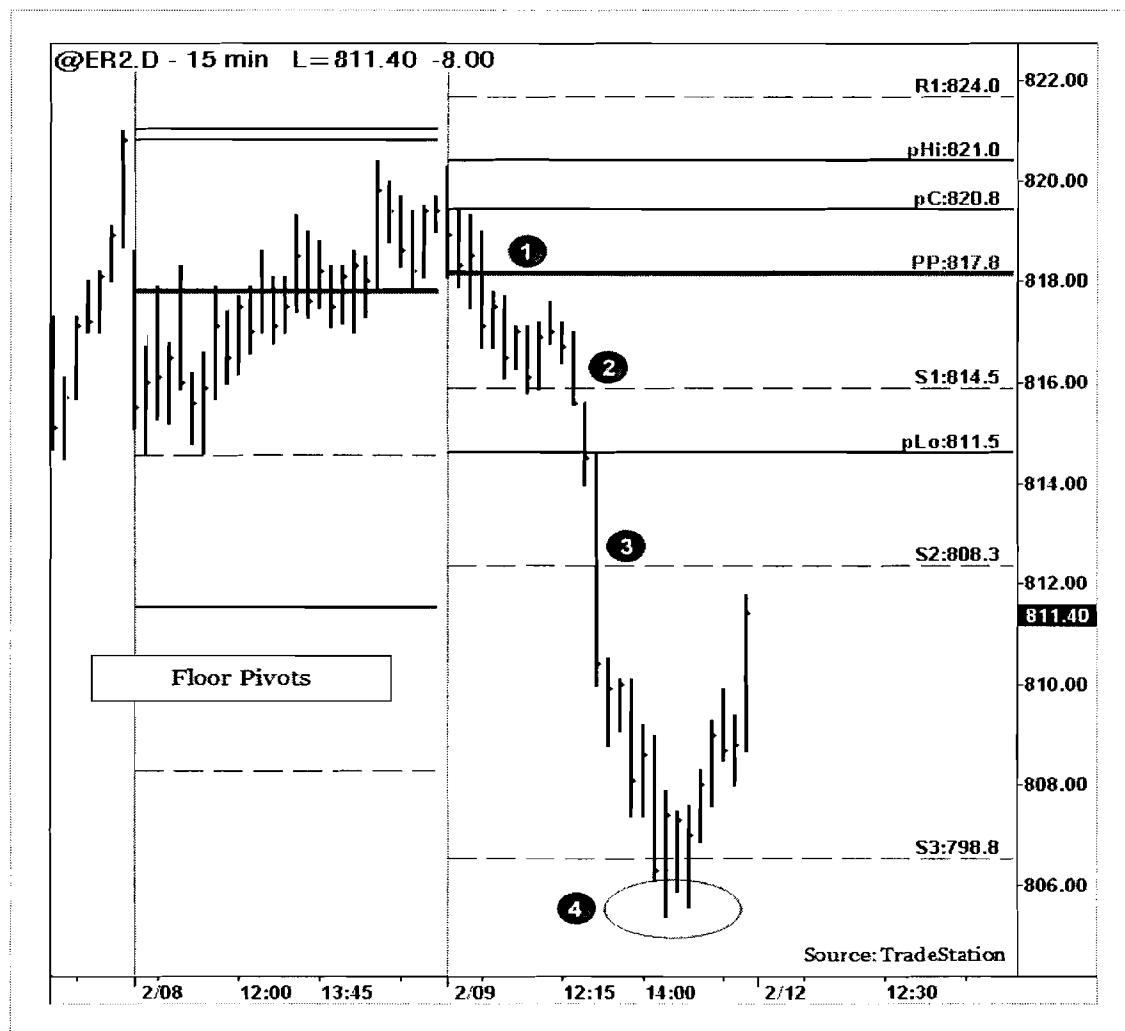
Third Support (S3) = $S1 - (H - L)$

Where, H=High, L = Low, C = Close

These equations above can be also applied to weekly, monthly Pivot calculations.



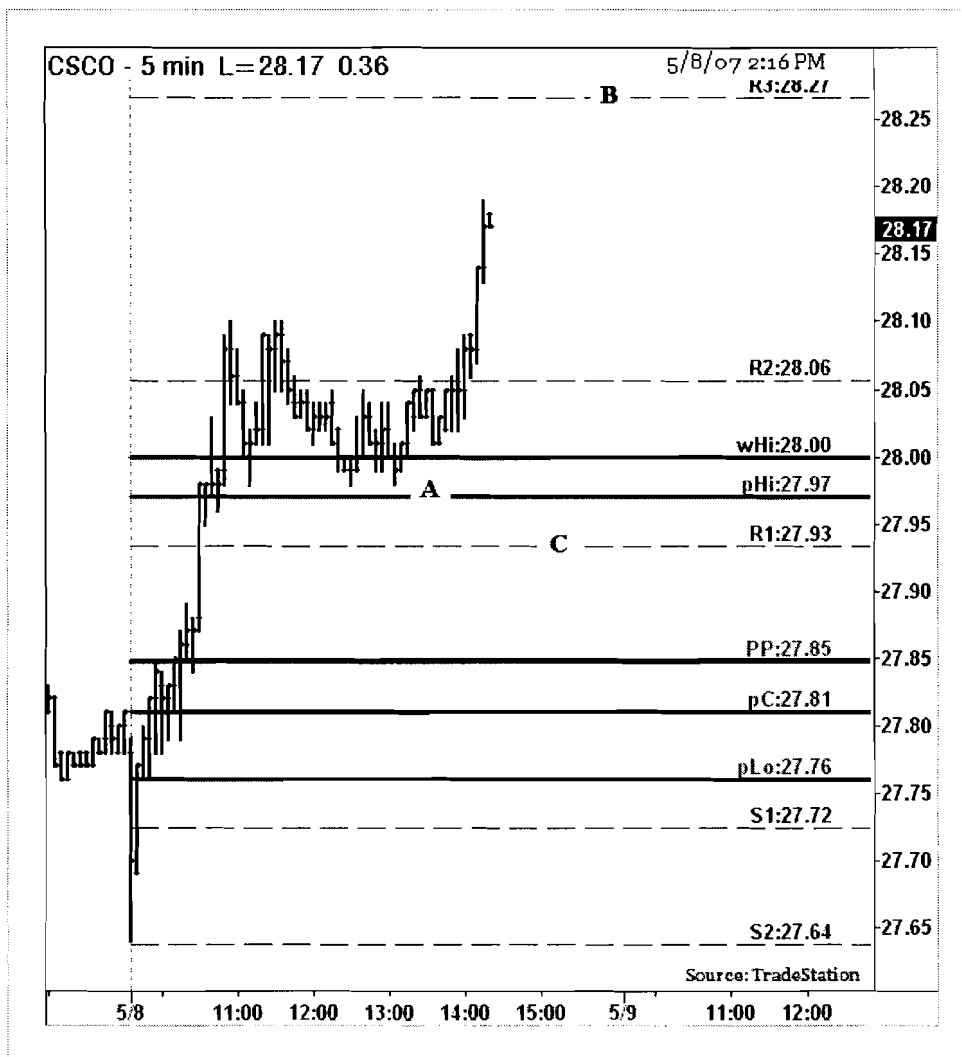
Trading Floor Pivots



Trading Floor Trader Pivots

The example above illustrates Pivot levels from the ER2 15m chart. On February 8th, the market traded in a narrow range around the Pivot level suggesting a contraction in the volatility. On the following day, ER2 opened above the previous close and traded near the previous day's high but could not hold on to this high prices. After the first hour, prices traded below the Pivot level at 817.8 suggesting a potential volatility/range expansion. ER2 continued to trade below the Pivot point, but above the key support level S1 until lunch time. At around 12:30p.m., ER2 broke through the key support level S1 at 814.5. A "short" trade was entered below the S1 level. The potential targets were the previous day's low at S2 levels. The volatility expansion was very strong as the market fell through S2 at 808.3 level suggesting a potential target S3 at 798. At around 2pm, the market stabilized at S3 level and attempted a rally back towards S2 level.

Trading Floor Pivots



Trading Floor Pivots

The chart above shows Floor Pivots from the Cisco (CSCO) 5 minute chart. On May 8th, in anticipation of CSCO's earnings report, the company rallied above the pivot level. A trade setup was presented when CSCO traded above the weekly high, but below the second resistance at R2. CSCO prices traded above the previous day's high at 27.97 and Pivot at 27.85 to suggest a potential bullish trend. At around 2 pm, CSCO closed above R2 to present a low-risk day trading opportunity.

1. Enter a "long" trade 5 cents above resistance at R2.
2. Place a "stop" loss at the previous high (at \$27.97).
3. Place a "target" at the third resistance R3 (at \$28.27).

3.2. Globex (Overnight) Pivots

Globex (Overnight) Pivots

Pit traders use Globex or overnight trading information extensively to determine key market trading levels for the current trading day. Globex trading hours start after the regular markets close and end before the regular markets open. For example, the Russell 2000 EMini futures on Chicago Mercantile Exchange (CME), the regular market hours are from 9:30 am ET to 4:15 pm ET. Globex trading hours for Russell 2000 futures starts at 4:30 pm ET and ends at 9:29 am ET. Similarly, the Dow Emini futures on Chicago Board of Trade (CBOT), has regular trading hours from 9:30 am ET to 5:00 pm ET. The Globex trading hours for Dow Emini futures are between 5:01 pm ET to 9:30 am ET.

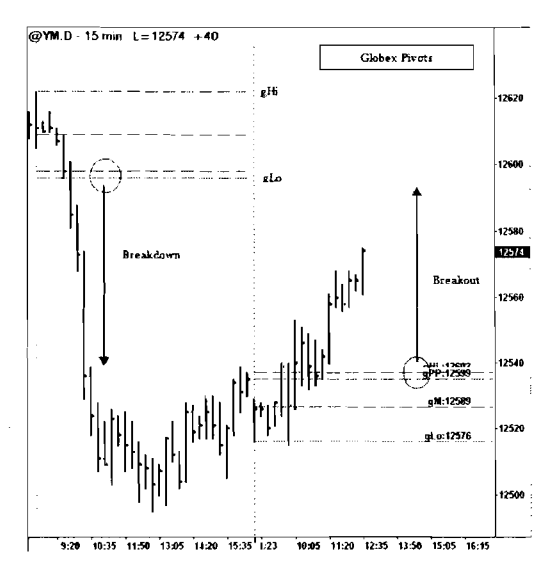
Unlike floor pivots, most traders use only Globex High, Globex Low and Globex Pivot values. Globex High, Low and Close levels are used to compute the pivot price level. The Globex pivot value and the high and low levels are used for measuring market strength and weakness. During most trading days, the Globex pivots may be valid until noon as traders still remember the Globex high and low levels. They are very effective in the first hour of market open. Like regular pivot levels, Globex highs and Globex lows are used for resistance and support levels. The first hour test of these levels and reversals from Globex highs and Globex lows are significant in intra-day trading.

Globex Pivots

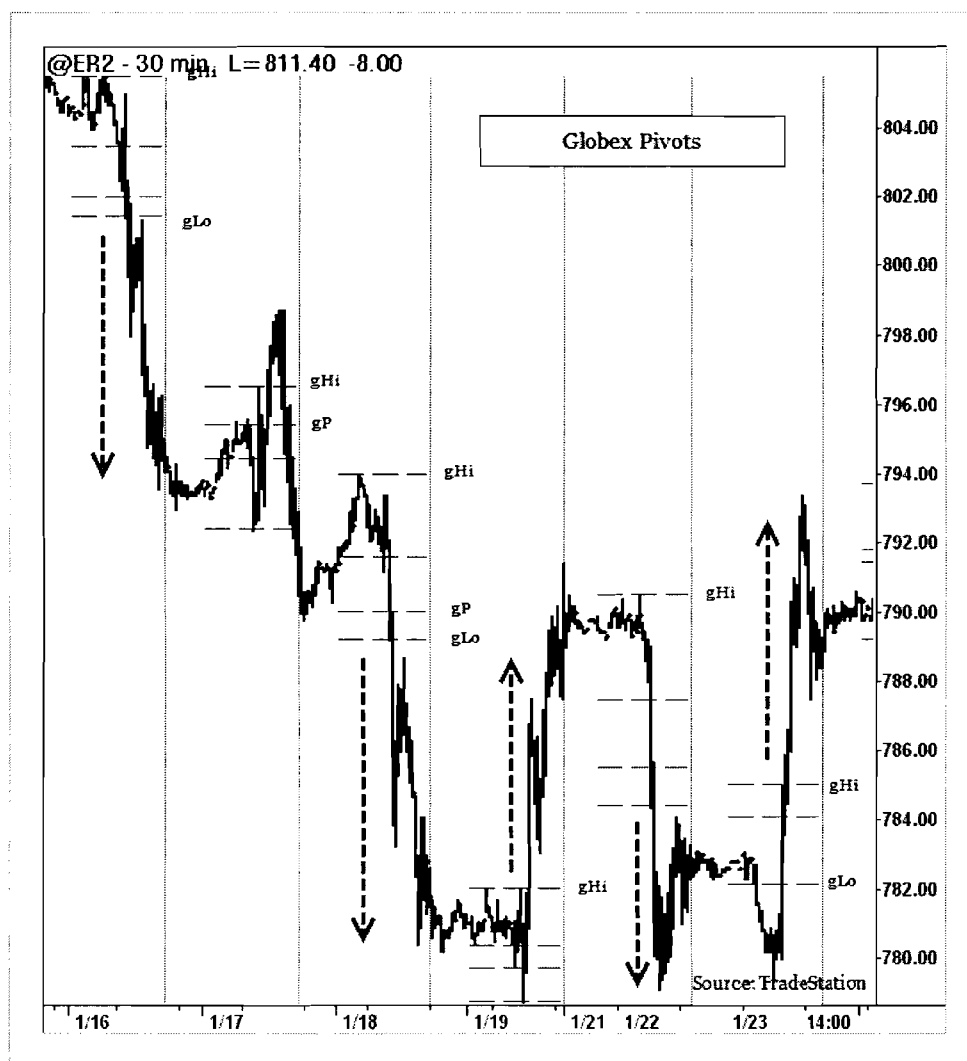
$$\text{Pivot Point} = (H+L+C)/3$$

Where,

- H = High of Globex Range
- L = Low of Globex Range
- C = Close of Globex Range



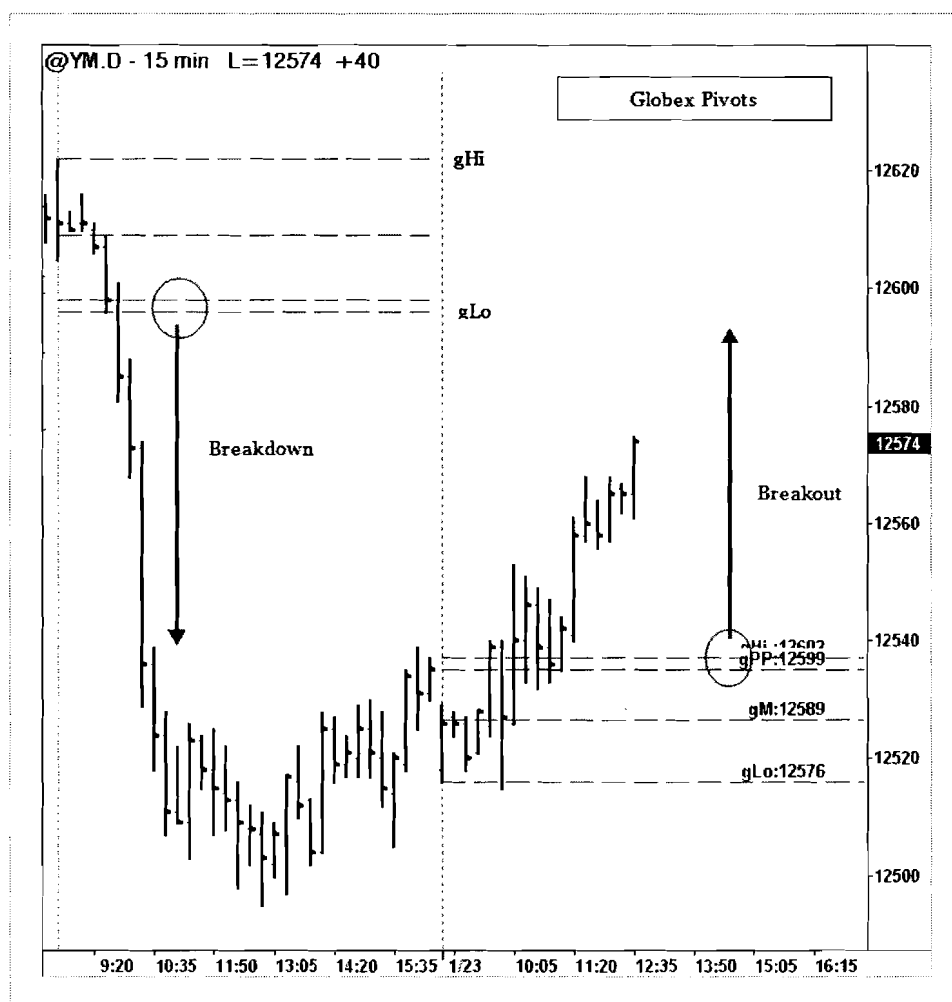
Trading Globex (Overnight) Pivots



Trading Globex Pivots

Globex pivots are for the overnight trading range. Globex pivot levels act as key resistance and support levels. They are mostly effective during the first few hours (until lunch hour) of trading. The example above shows the Russell Emini futures (ER2) from the 30 minute chart (24 hour chart). Globex high, low, Globex midpoint and Globex pivots are plotted on the chart. Arrows are placed on the chart to highlight the breakouts and breakdown from Globex high and low levels during the intra-day trading.

Trading Globex (Overnight) Pivots



Trading Globex Pivots

The chart above shows Globex pivot values from the Dow (YM) Emini Futures 15 minute chart. The chart shows January 22nd and January 23rd Globex pivots. On most trading days, when markets trade through Globex lows, it strongly suggests a weaker market, and when markets trade above the Globex highs it suggests a stronger market. On January 22nd, YM traded below the Globex low during the first hour at 12600 and traded all the way to a low of 12500. On January 23rd, YM traded above the Globex high during the first hour (above Globex pivot 12540) suggesting higher values for the rest of the day.

3.3. Opening Range Pivots

Opening Range Pivots

The Opening Range time is the initial time frame of trading for a stock or commodity at the start of each new trading session. For day-traders, this range could be the first 15 or 30 minutes and for swing or position trades it is the first hour range. Once this range is identified, the first hour pivots and support/resistance levels are calculated. This range usually sets the direction for the day and acts as important levels for intra-day trading.

Trading using the Opening Range pivot produces excellent results. In stronger markets, a long trade is initiated when prices pullback to the Opening Range pivot. A target for this trade is set at the first resistance. In weaker markets, prices crossing below the Opening Range pivot could signal a trend reversal and a short trade may be initiated.

Confluence of daily pivot levels and first hour Opening Range Pivots could be significant in trading. Most days, the first hour of highs and lows clearly signal the market's strength. Trading below the pivot and the first hour lows set the market in a bearish mode. Similarly, trading above the first hour high and above the first hour pivot sets the market in a bullish mode. The advantage of using Opening Range pivots is market stability. In the first 30-60 minutes (the amateur hour) of the market open, prices go through a series of gyrations to settle on a market direction for the rest of the day. Most seasoned traders wait for prices to pick a clear market direction with the opening range (the first hour) data before start trading.

Opening Range Pivot Calculations

Pivot Point (PP) = $(H+L+C)/3$

First Resistance (R1) = $2*PP - L$

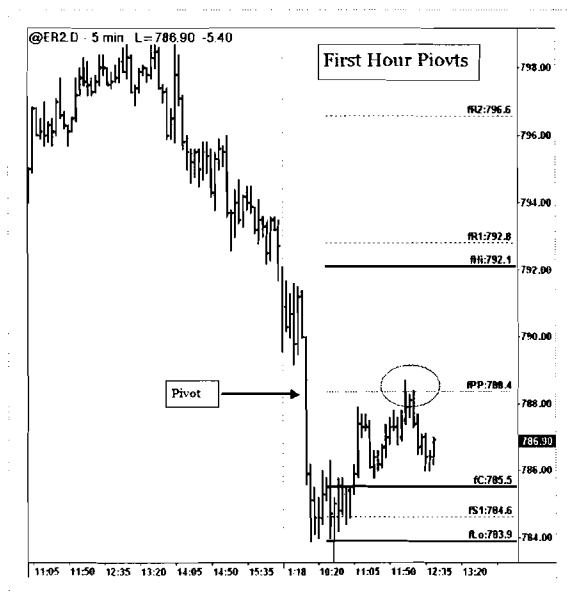
First Support (S1) = $2*PP - H$

Second Resistance (R2) = $PP + (R1-S1)$

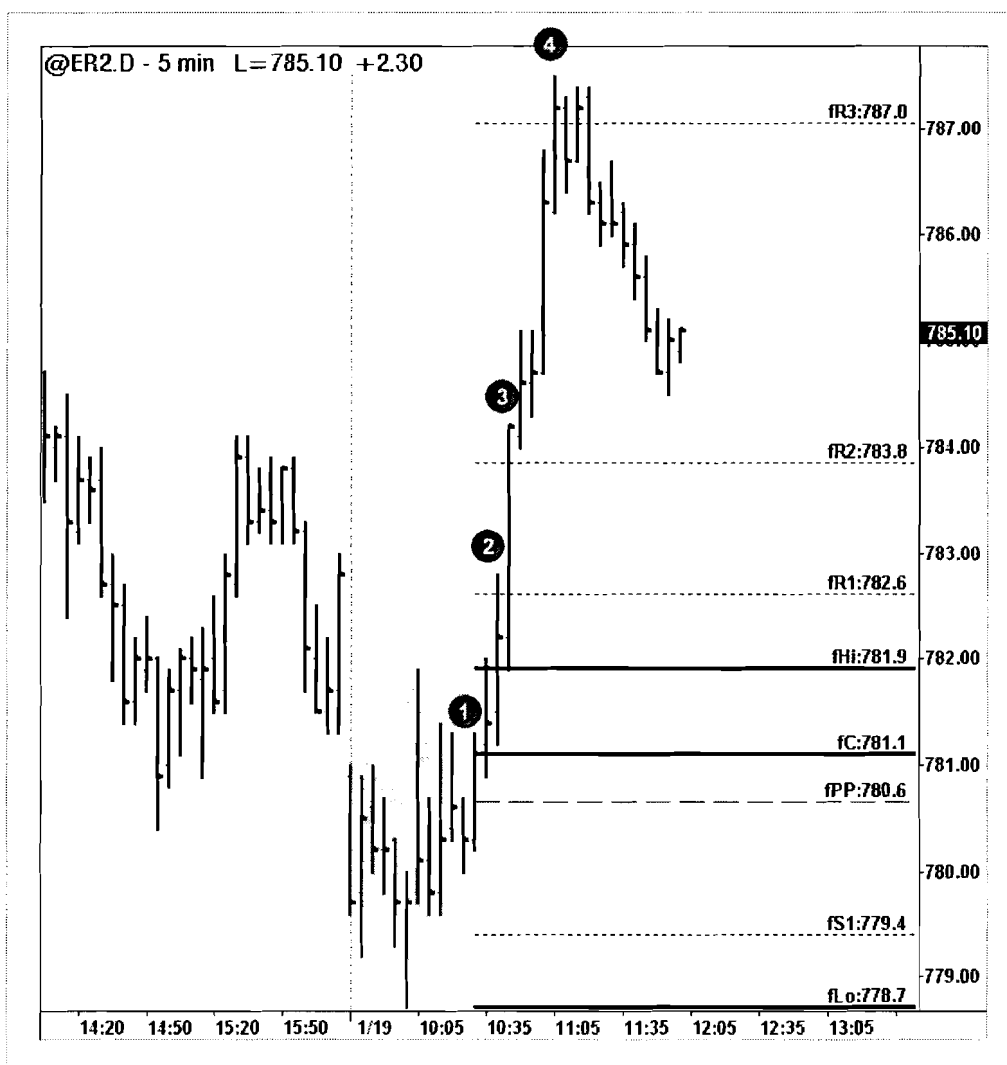
Second Support (S2) = $PP - (R1-S1)$

Third Resistance (R3) = $R1 + (H - L)$

Third Support (S3) = $S1 - (H - L)$



Trading Opening Range Pivots

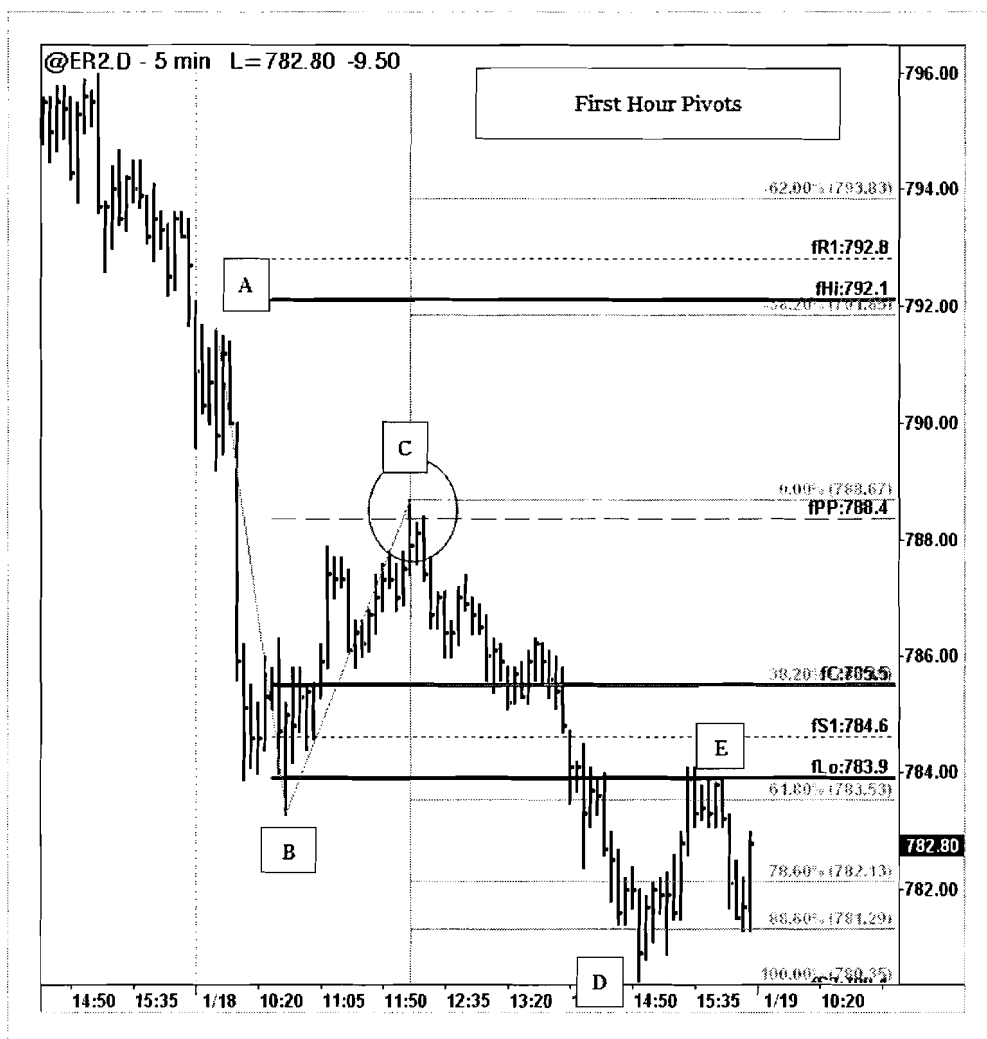


Trading Opening Range (First Hour)

The example above illustrates the Opening Range pivots from the ER2 5m chart. On January 19, 2007, ER2 traded in a narrow range in the first hour-Opening Range (shaded area) from 778.7 to 781.9 and closed at 781.1 (at marker 1). After the first hour of trading, ER2 set to trade above the Opening Range high at 781.9 and the Opening Range pivot at 780.6 signaling an upside day with range expansion.

1. Enter a "long" trade above the first hour high at 781.9.
2. Place targets at R2 and R3 levels.
3. Place a "stop" order few ticks below the pivot at 780.6.

Trading Opening Range Pivots



Trading Opening Range (First Hour) with ABC Pattern

The chart above shows Opening Range pivots along with an ABC pattern trade. On January 18, 2007, ER2 opened at a high of 791.8 and drifted lower in the first hour to trade near 784 area. After the first hour of trading, ER2 attempted to rally to the first hour pivot at 788.4 to form an ABC pattern. The trend and market direction was weak as prices continued to trade below the first hour pivot and the first hour high. Prices attempted to rally, but around noon, ER2 could not close above the first hour pivot and reversed its direction to signal a potential short trade.

1. Enter a "short" below the first hour pivot at 787.8.
2. Place a "stop" order above the first hour pivot at 788.5.
3. Place a target from 78.6% to 100% of AB range (from C) to 781.3.

3.4. FibZone Pivots

Fibonacci Zone Pivots

John T. Jackson described a concept called “Zone Probability Pattern Analysis,” in his book *Detecting High Profit Day Traders in the Futures Markets*. In his book, he describes the dynamic statistical analysis of “Fibonacci Zones” using Open, High and Close to find high percentage support and resistance zones for the entire trading day.

Robert Krausz has completed extensive analysis on “High Probability of the Fibonacci Zones” (HPFZ) concept by using “Probability Matrix.” Krausz used various close and open combinations as price moves from zone to zone to achieve the highest probable Open to Low combination for the day.

In this chapter, I present how I use Fibonacci Zone Pivots for my trading. I use HPFZ zone concept as boundaries for trading key support and resistance areas. For additional analysis, please read J.T. Jackson’s book on Zone Pattern Probability Analysis (See References)

Computing FibZone pivot starts with calculating the pivot $(High + Low + Close) / 3$ for the current day and then projecting the “zone pivots” for next day. Adding daily range and its multiples to this pivot point gives various potential resistance and support levels. The boundaries of these resistance/support levels are marked as FibZones. An extensive study has been done on the relationship between yesterday’s close and today’s open within these zones to find a potential daily “close” for the current day. I use FibZones in my personal trading and have found them to be valid and useful. My usage of HPFZ is limited to finding intra day resistance and support areas and not for the zone analysis.

FibZone Pivots

Pivot Point (PP) = $(H + L + C) / 3$

Daily Range (DR) = $(H - L)$

First Resistance (R1) = $PP + 0.5 * DR$

Second Resistance (R2) = $PP + DR$

First Support (S1) = $PP - 0.5 * DR$

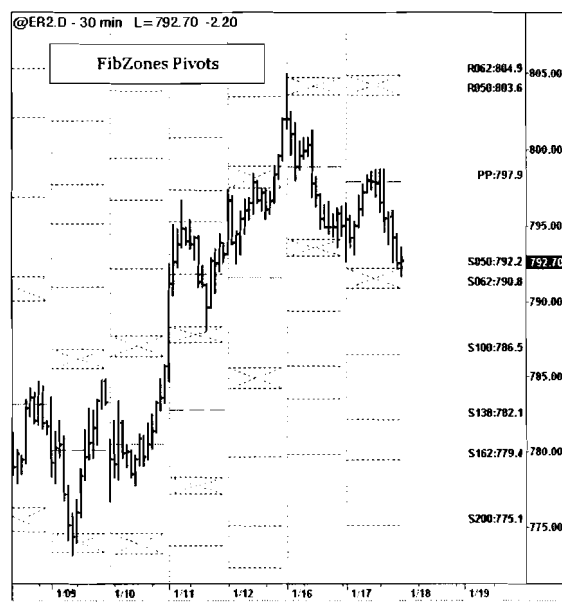
Second Resistance (S2) = $PP - DR$

Resistance Band (RB1) = $PP + 0.618 * DR$

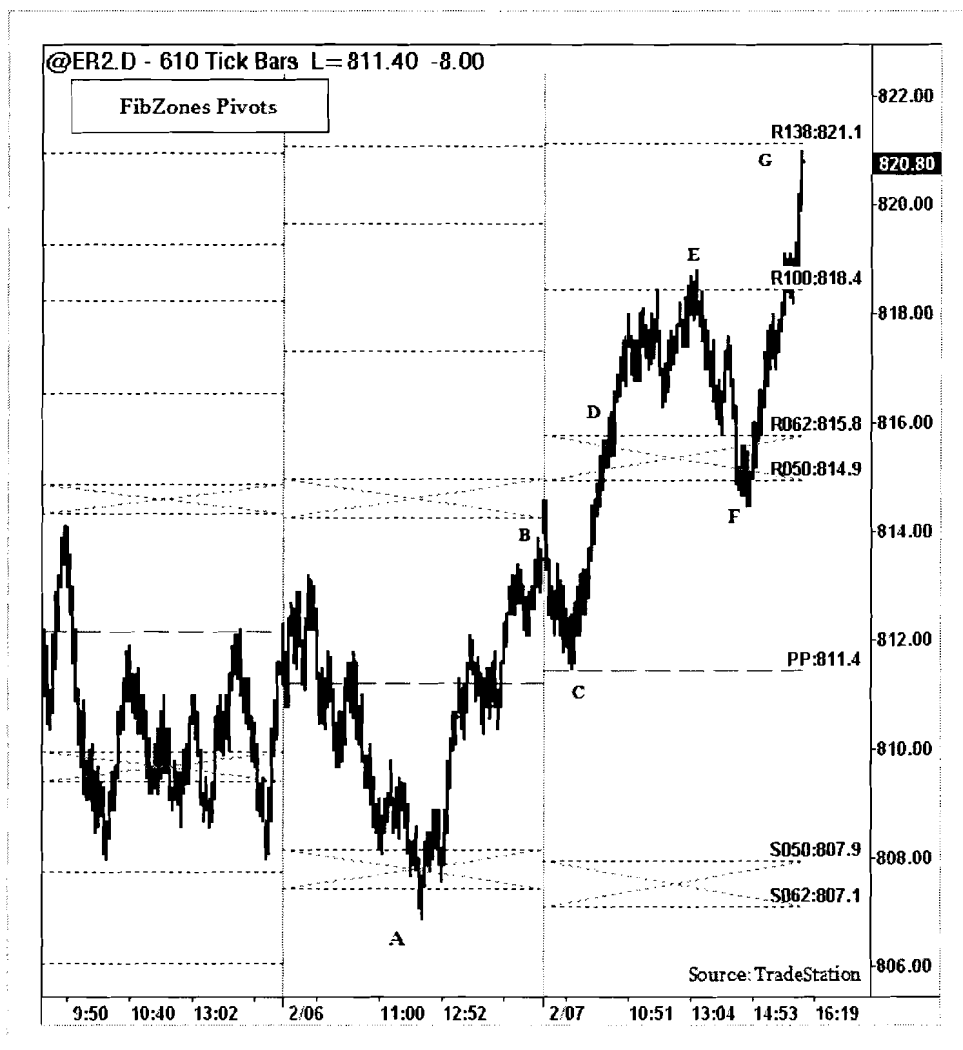
Support Band (SB1) = $PP - 0.618 * DR$

Resistance Band (RB2) = $PP + 1.382 * DR$

Support Band (SB2) = $PP - 1.382 * DR$



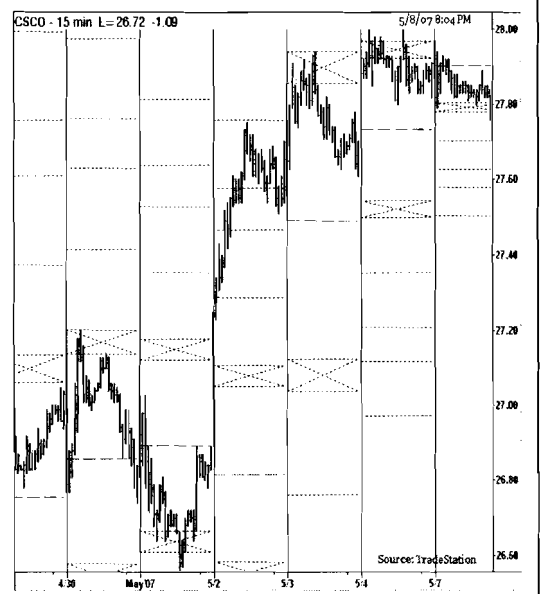
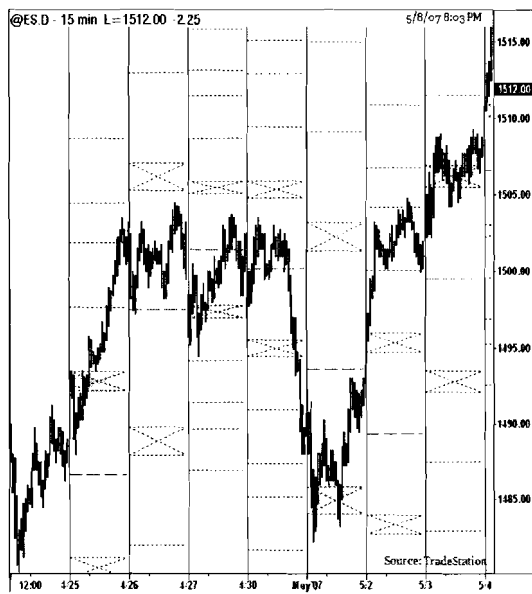
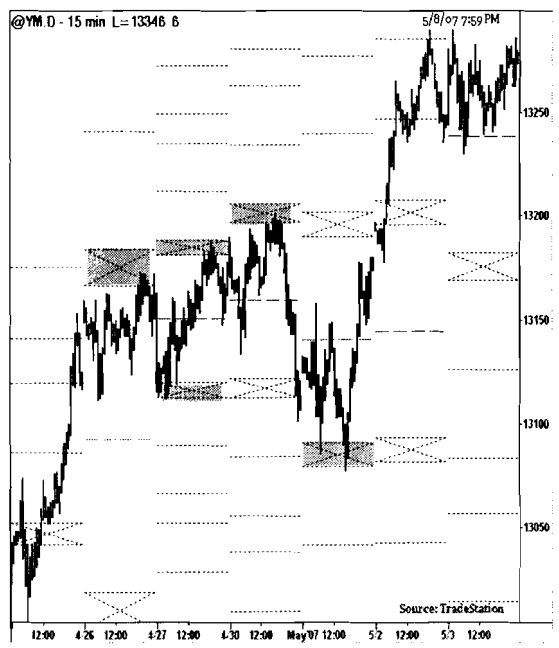
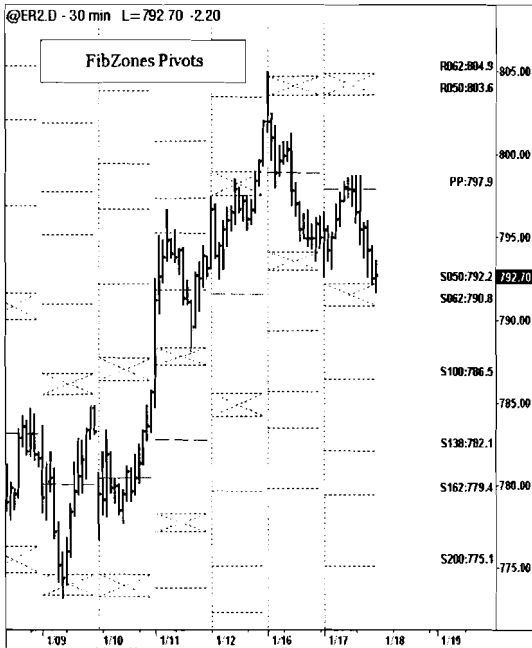
Trading Fibonacci Zone Pivots



Trading FibZone Pivots

The example above shows FibZone pivots plotted from the Russell Emini futures (ER2) 610 chart. Fibzones are plotted at the end of the day for the next trading day. On February 6th, ER2 sold-off and found support near the “support” band between the S50 to S62 range. In the afternoon, ER2 rallied back from “support band” (A) and closed in the “resistance band” (B). The following day, ER2 had a brief sell-off to the pivot point (C) and rallied to the “resistance” band (D). The first test of resistance to the rally came in the afternoon at 100% of range test E. A pullback to the “resistance” band-R50 to R62 is expected after the morning rally to (F). Another continuation of the rally, sent prices to 138% of previous day range above pivot level (G).

Trading Fibonacci Zone Pivots



Chapter 4: Fibonacci

4.1. Fibonacci Trading

Fibonacci Methods

Fibonacci numbers are pervasive in the universe and were originally derived by Leonardo Fibonacci. The basic Fibonacci ratio or “Fib ratio” is the Golden Ratio (1.618). Fibonacci Numbers are a sequence of numbers where each number is the sum of the previous two numbers.

The series of Fib Numbers begin as follows: 1,1,2,3,5,8,13,21,34,55,89,144,233,317,610....

There are plenty of materials and books about the theory of how these numbers exist in nature and in the financial world. A list of the most important Fib ratios in the financial world which are derived by squaring, square-roots and reciprocating the actual Fibonacci Numbers are depicted below:

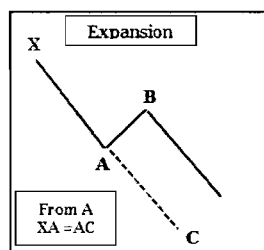
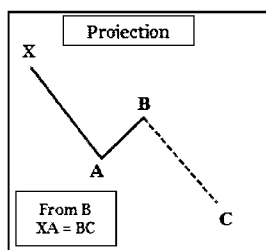
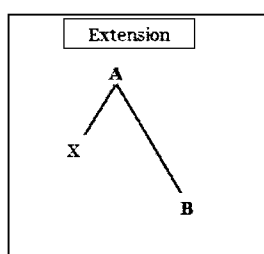
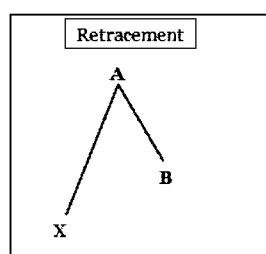
Key Set of Fibonacci Derived Ratios in Trading are:

0.382, 0.500, .618, 0.786, 1.0, 1.272, 1.618, 2.0, 2.62, 3.62, 4.62

Secondary Set of Fibonacci Derived Ratios in Trading are:

0.236, 0.486, 0.886, 1.13, 2.236, 3.14, 4.236

Most trading software packages have Fibonacci drawing tools which can show Fib retracements, Fib Extensions and Fib Projections. In addition. Fib Numbers are also applied to “time” and to “price” in trading.



Types of Fibonacci

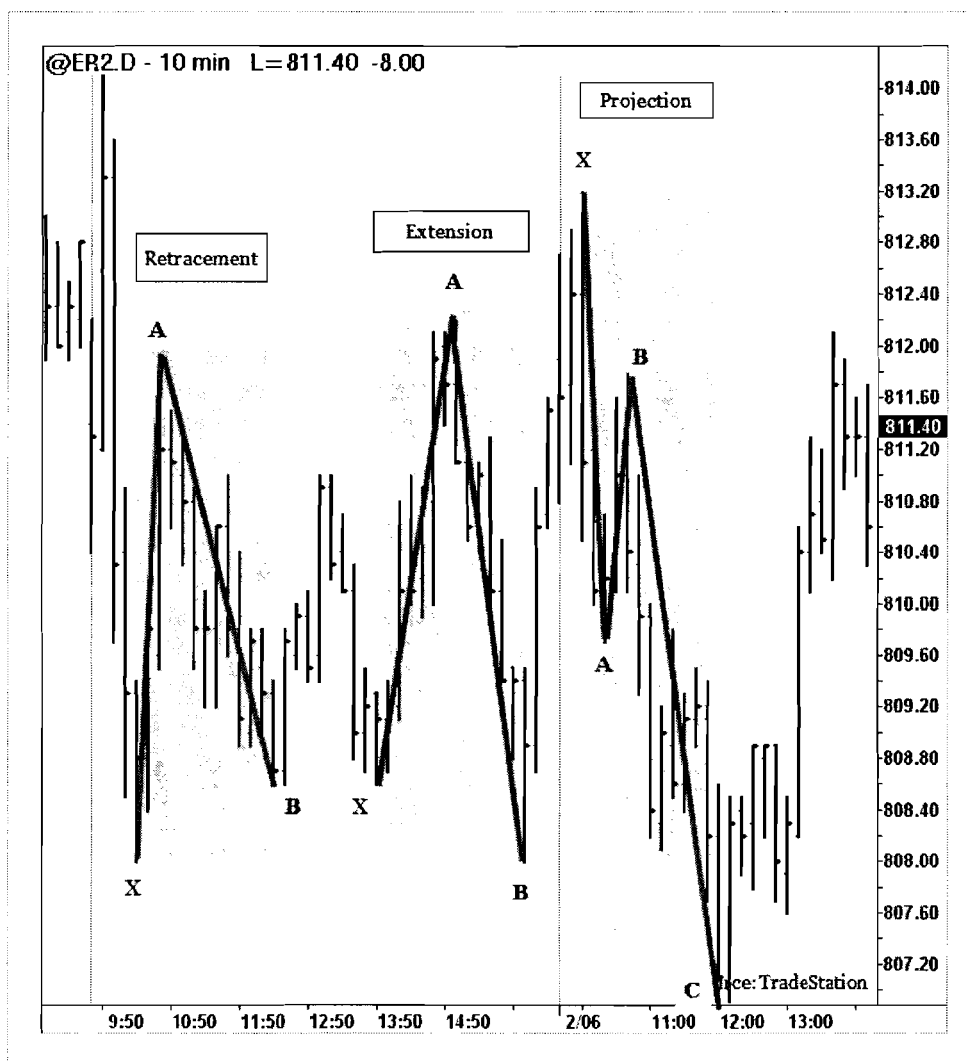
Retracement: From the swing XA, a Fibonacci ratio length is retraced to B.

Extension: From the swing XA, more than 100% of the swing XA is extended from A to B.

Projection: From the swing XA, a retracement is made to form the AB leg. The swing XA is projected from B to C.

Expansion: From the swing XA, a retracement is made at B to form AB. A projection is plotted from A in the same direction and length of XA to C.

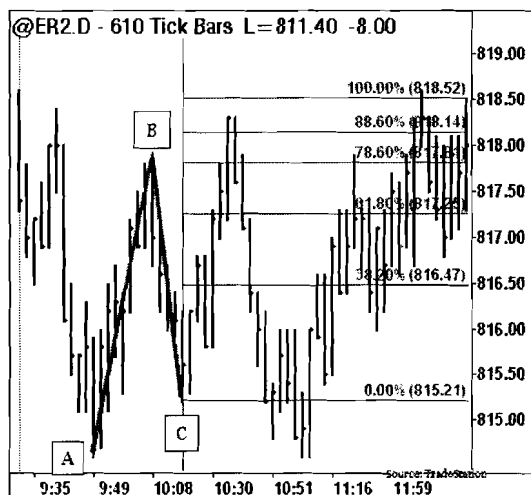
Fibonacci Trading



Fibonacci Trading

The chart above illustrates Fibonacci trading from the Russell Emini futures (ER2) 10 minute chart. First, a retracement is shown in the shaded area for XA to AB. Secondly, an extension of XA to AB is shown. Thirdly, the projection of the XA length is made from B to C. In the forgoing projection example, B was retraced to .62 of XA and then a projection of 1.27 to 1.62 of XA swing is expected. C was formed at 1.38 of XA.

Fibonacci Drawing Tools



Retracement and Extension Drawing Tools

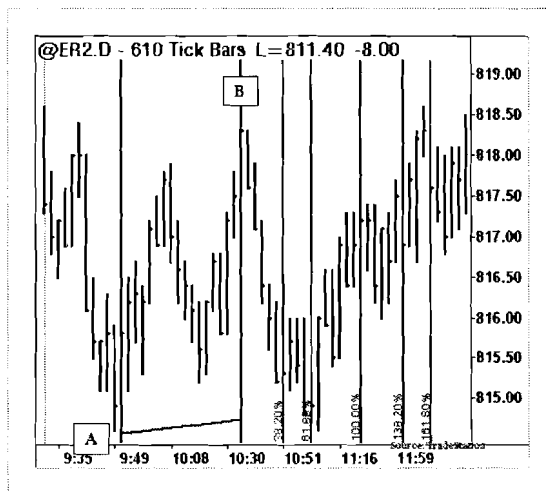
Most software packages have Fib retracement and extension drawing tools. These tools allow a user to pick a “swing high” and “swing low” and draw retracement and expansion levels. The Fib levels can be also customized for various Fib Numbers.

Another Fib drawing tool is an extension tool. This tool lets a user pick three different swing points A, B and C and then draw “swing” extensions from C. This is a very valuable tool since it plots fib. levels from the trade level C.

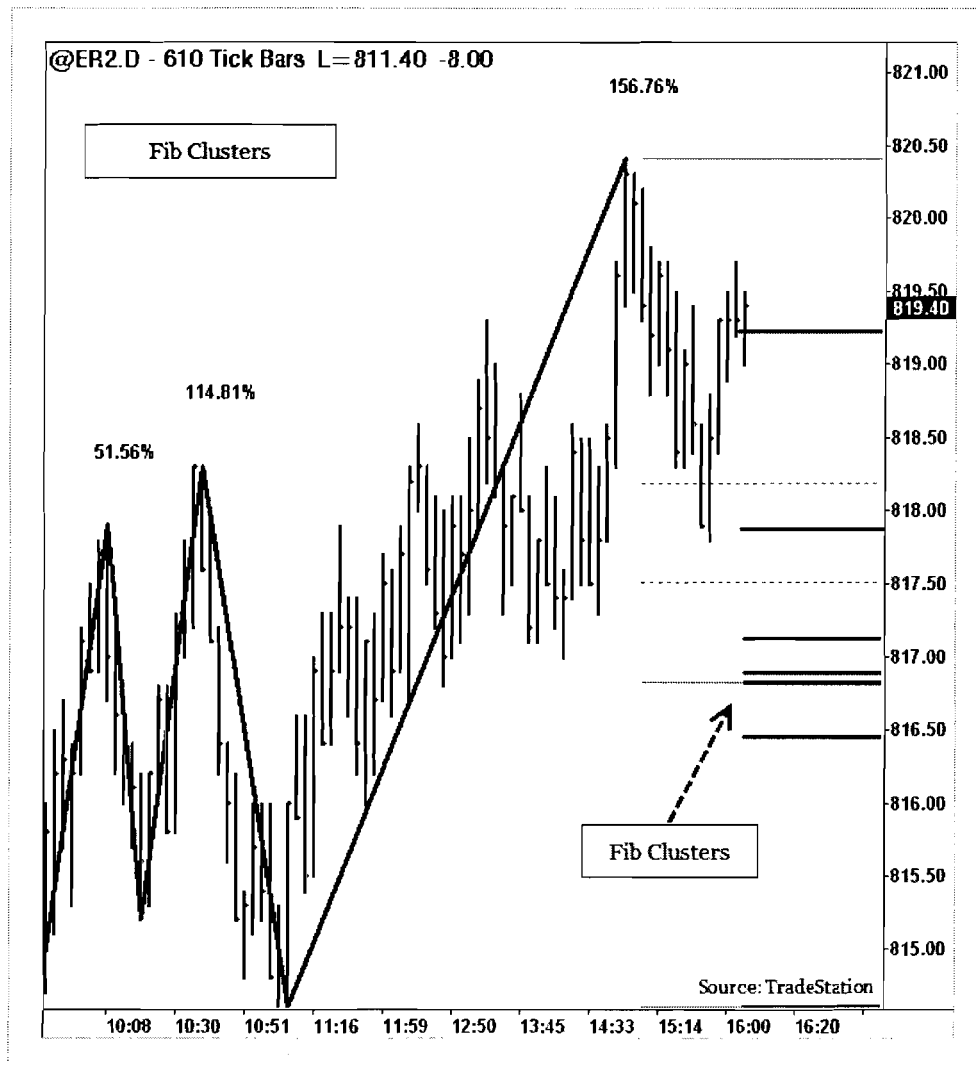
Fibonacci Time Extension Tool

Fibonacci extensions can also be measured for “time” levels. Most software packages also provide a Time Extension tool.

This tool helps find key turning points using “Fib Time.” It allows a user to pick a “swing low” at A and a “swing high” at B and plot the Fib Time extensions from this time-frame. The projected time levels provide the user with potential “swing” turn points.



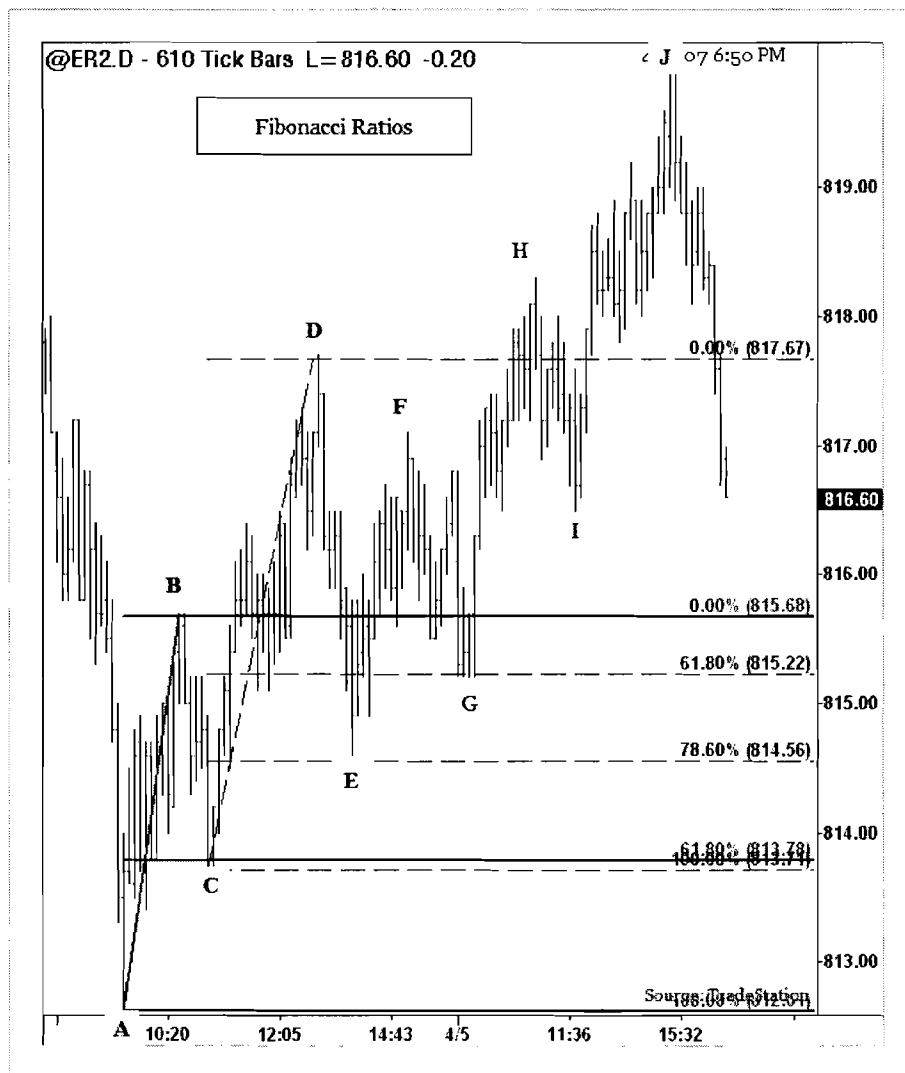
Fibonacci Clusters



Fibonacci Clusters

A confluence of Fib. retracement numbers signifies a stronger area of support and resistance. The confluence of multiple Fibonacci retracements in a fairly tight area are computed using prior swings both retracement and extensions with certain criteria. Then each Fib retracement/extension level is grouped to generate a confluence within some threshold to find Fibonacci clusters in an area. These "cluster" levels are more significant than a single fib retracement itself. The trades are either initiated or closed at these confluence levels.

Trading Fibonacci Retracement Levels



Trading Fibonacci Retracement Levels

The chart above illustrates an example of trading Fibonacci ratio levels from the Russell Emini futures (ER2) 610 tick chart. One of the effective method to trade Fibonacci levels is to trade pullbacks in the direction of its primary trend near the Fibonacci retracement levels. For the first “swing” at AB, trade reversal occurred at the 61.8% retracement level. A “long” trade is entered at C with a “stop” order placed below A. On the second “swing” at CD, a “long” trade is entered at the 61.8% retracement level. The swing retraced to E to 78.6%, but rallied back into DE swing. For the second trade, a “stop” order is placed at C. Targets are set at the top of “swing highs” at A, D, F and H.

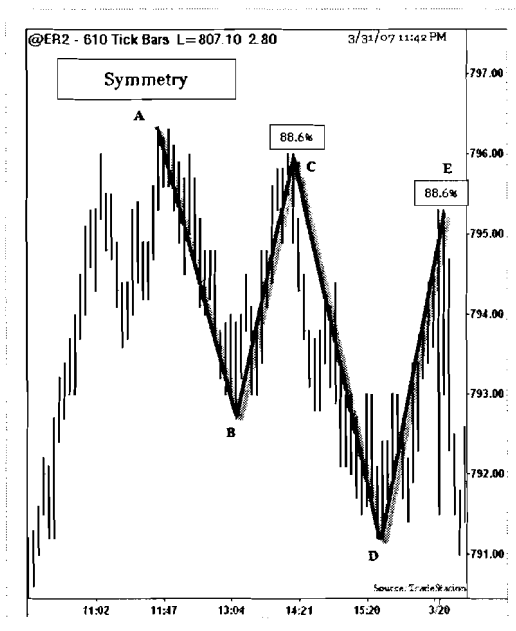
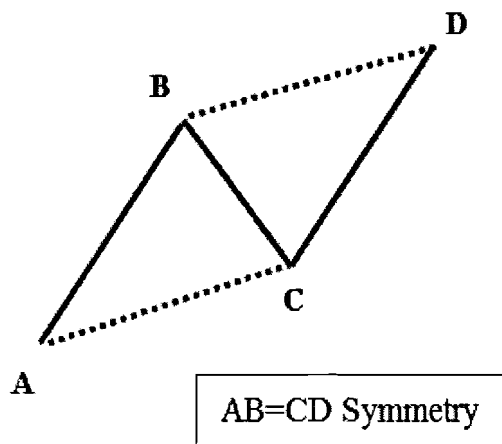
4.2. Symmetry Patterns

Symmetry Patterns

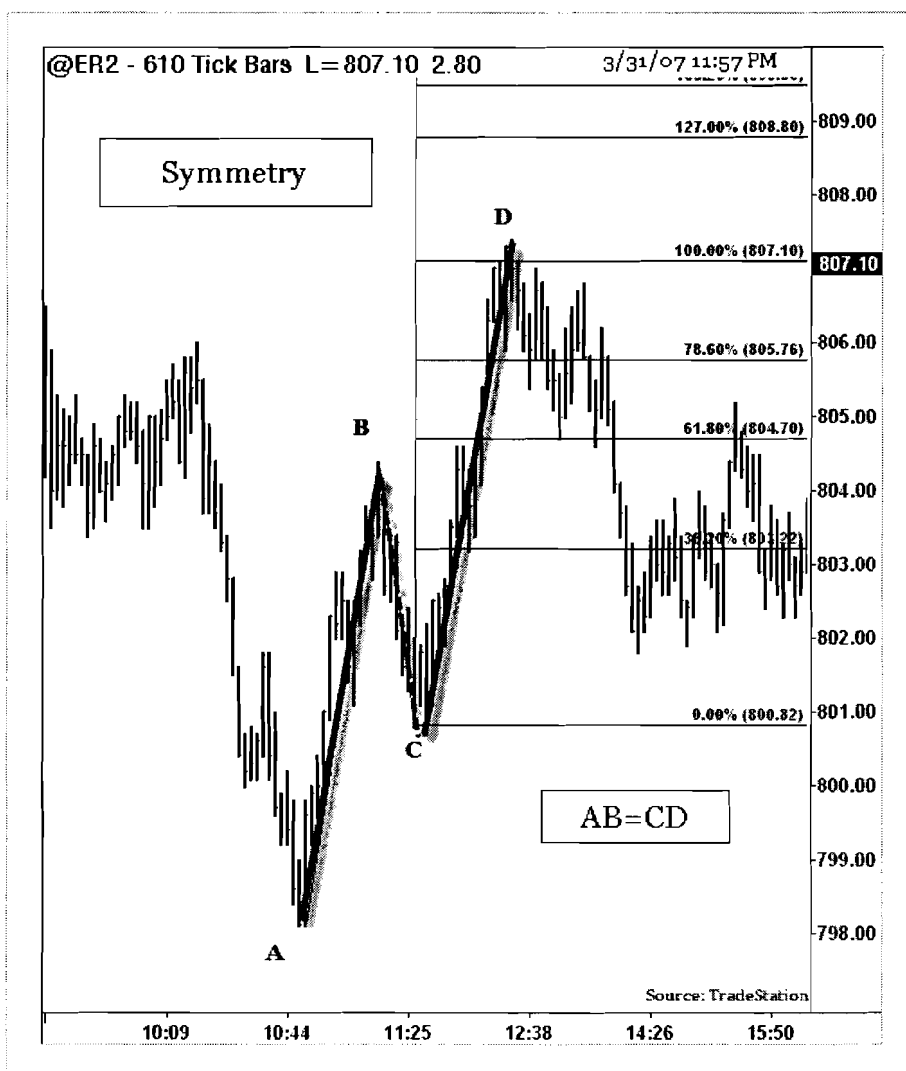
Markets demonstrate repetitive patterns where prices oscillate between one set of price ratios to another making price projections possible. Market trends can be defined by geometric relationships as they exhibit harmonic relationships between the price and time swings. Markets also form “cycles” around the price and time levels. Many investors/traders use “cycles” and “harmonic” relationships to project future swing prices and times.

“Symmetry” is visible in all markets and in all time-frames. “Symmetric” rallies and declines give traders an advantage to determine the key turning points. A cluster of similar extensions and similar retracements at key price ranges, or some important levels provide insights into future significant resistance and support levels. In addition to knowing key turning points, the benefits of trading symmetric “price/time” cluster levels include low-risk trades.

Gann, Fibonacci and Elliott all have studied market symmetry and found valid theories. These patterns exist in all forms in nature and certainly exist in the markets. One of the best ways to confirm “Symmetry” in the markets is to check “price” and “time” using two or more cluster confirmations. Another key method to compute these patterns is to use “percentage change of price” between market “highs” and market “lows.” “Symmetry” is a science by itself and traders take great advantage of knowing the potential turning points/levels using these methods.



Trading Symmetry Patterns

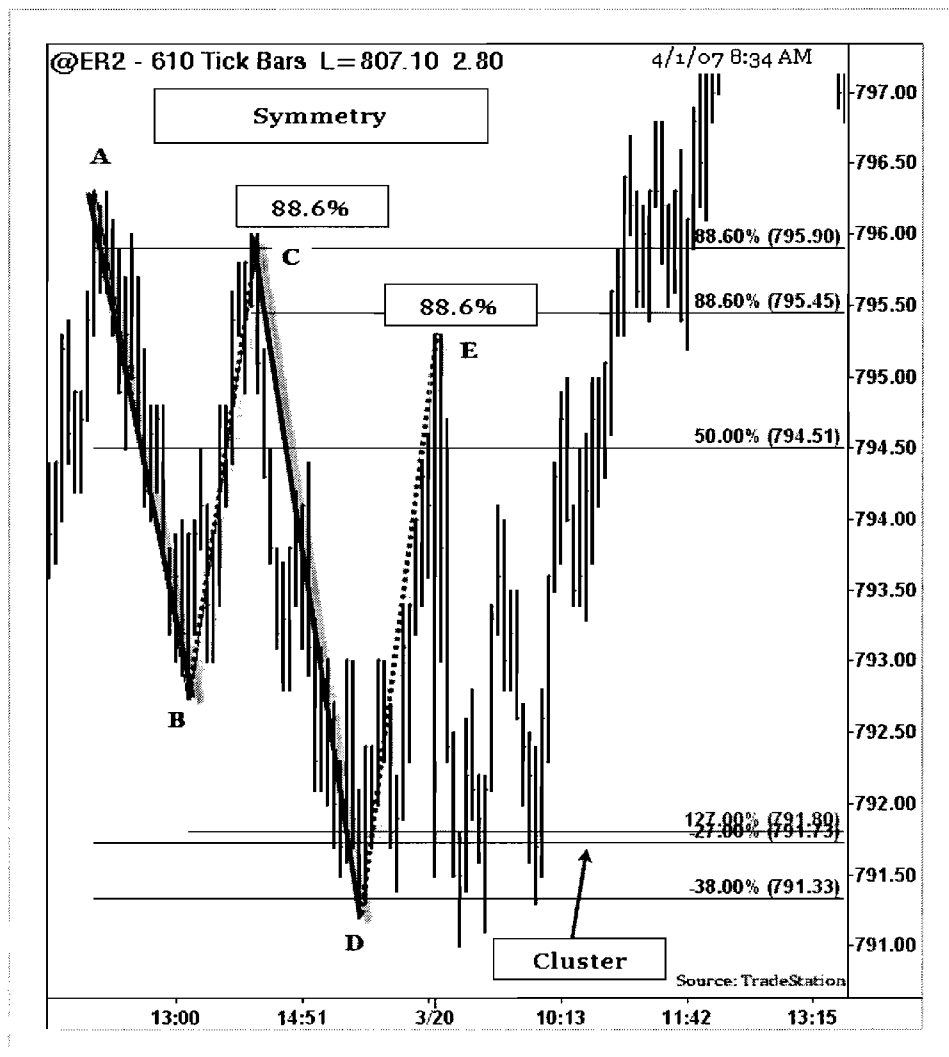


Trading Symmetry

The chart above illustrates the “Symmetry” pattern from the Russell 2000 futures (ER2) 610 tick chart. Symmetrical swing lengths are shown after a 50% retracement at BC swing. Market “symmetry” of 100% extension is expected after a retracement of less than 50% retracement levels. If the retracement exceeds 50%, the extension may be less than or equal to 100%.

1. After ‘BC’ retracement, enter a “long” trade one-tick above the B.
2. Place a “stop” order one-tick below the low of “C”.
3. Target 100% of the AB range from level C to D.
4. Another potential retracement is expected at D to 50% of the entire range of AD.

Trading Symmetry Patterns



Trading Symmetry

The example above shows “Symmetry” and cluster of harmonic ratios from the Russell 2000 (ER2) 610 tick chart. After BC retracement of 88.6%, a Symmetrical extension at CD is formed near the “cluster” zone. This zone is where two or more harmonic levels are grouped at a single level. The “cluster” is formed at the retracement of 127% to 138% of AB and 127% of BC swings. The zone also acts as key support and resistance areas for trading. After another retracement of DE at 88.6% from the “cluster” zone, ER2 rallied to new intra-day highs.

4.3. Market Fractals

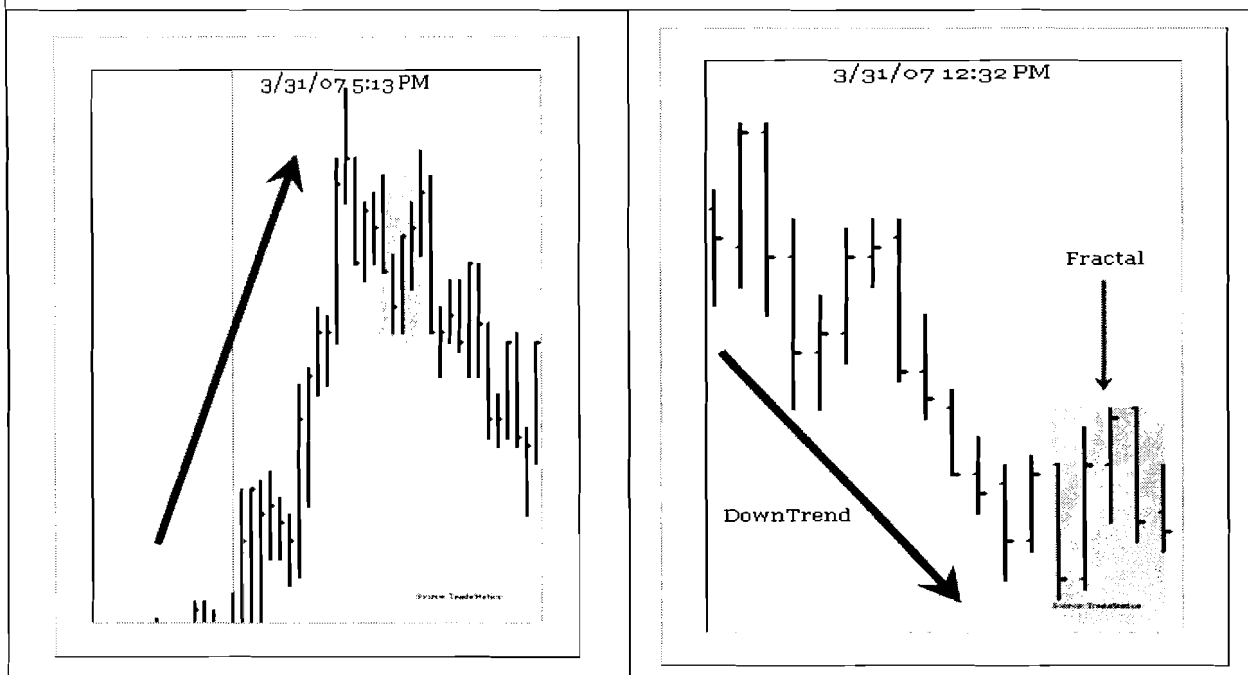
Market Fractals

Financial markets are Complex, Non-Linear and Chaotic. Chaos is the highest form of order and posses a highly deterministic behavior. These chaotic market behaviors are represented by graphical structures usually initiated by a pattern called "Market Fractal." Trader's decisions are usually decided by complex series of events and these events influence price changes in the markets. Many times, all these events are rhythmically synchronized with Momentum, Volume, Time and Price. Price is the last one in this series to be effected. Traders get significant benefits from knowing the beginning of market structures (Fractals). It's not easy to identify market "Fractals" but there may be a set of rules and patterns/events which may help traders to identify them.

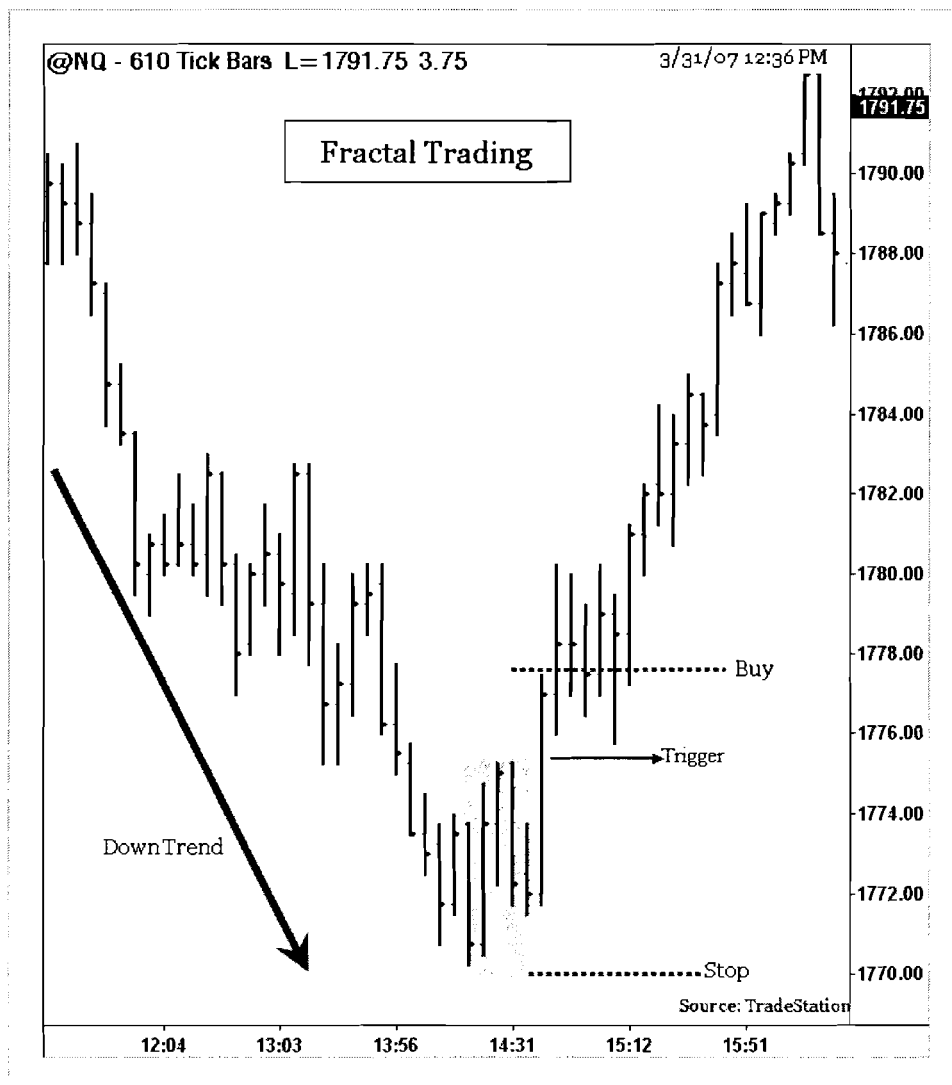
"Fractals" are similar patterns which repeat themselves. A beginning "Fractal" pattern resembles the overall pattern of the entire market structures. Fibonacci numbers and Elliott wave patterns were first initiated by "Fractals". On the expansion, Elliott waves consist of a series of "Fractal" structures.

In a simplest market form, as an example, a "Fractal" consists of 5 bars. After a prolonged downtrend in the markets, a 5-bar "Fractal" is formed to signal a potential change in the trend. This "Fractal" has three bars with higher highs and two bars with lower lows. Trades are initiated when another higher-high is formed after this 5-bar "Fractal". A bearish "Fractal" is the reverse of the above.

As in any technical indicator, "Fractals" form, fail, re-fail and re-form. "Fractals" work in all markets and in all time-frames. Fractal theory is very powerful, but it does need confirmation indicators such as Momentum, Divergence and price-action to be valid.



Trading Fractal Patterns

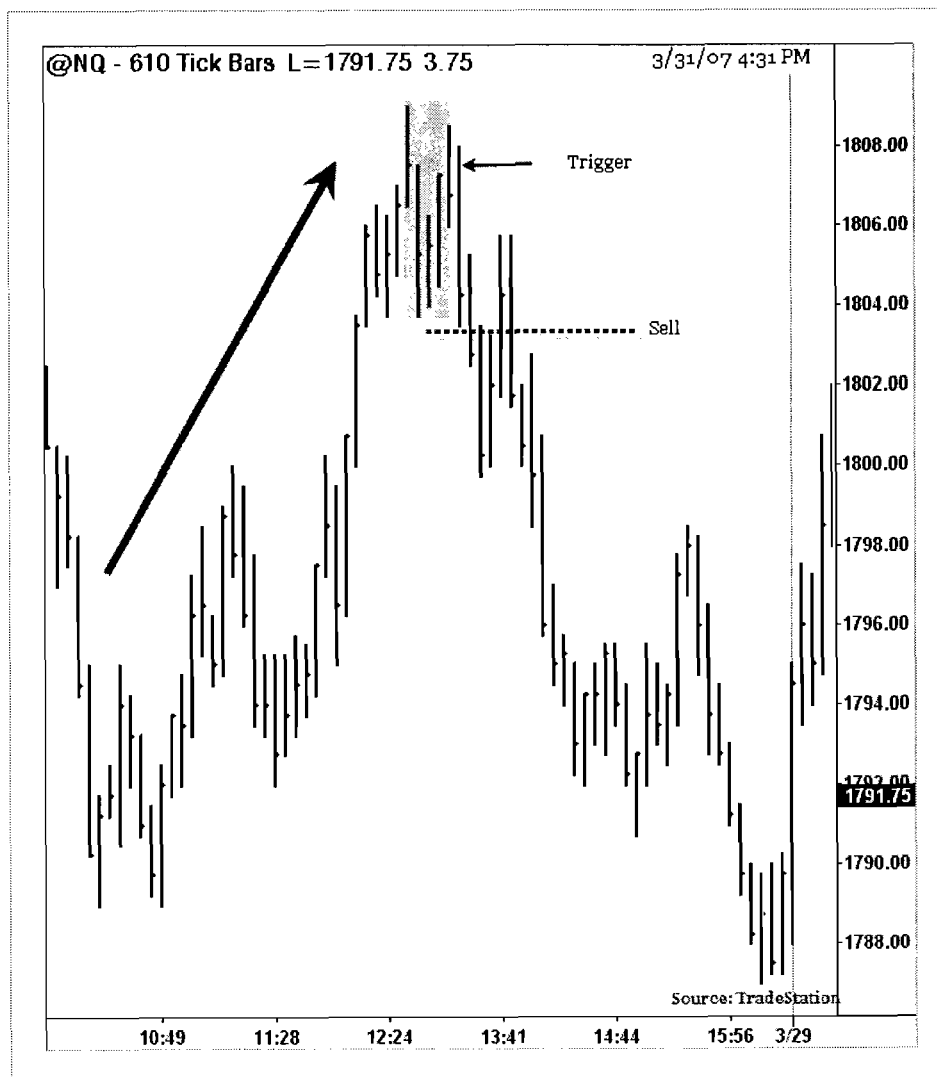


Trading Fractal Patterns

The example above shows a “Fractal” pattern formation from the Nasdaq Emini futures (NQ) 610 tick chart. On March 29, 2007, NQ futures were in a downtrend all day and lost over 20 points. At about 2:30pm, NQ made a series of higher-highs followed by two lower-low bars to form a “Fractal.” This “Fractal” suggests that a down-trend may be over and a significant trend change may be in the works.

1. After a 5-bar “Fractal” formation, wait for a bar to close above the previous bar’s high and enter a “long” trade above the previous bar’s high.
2. Enter a “stop” loss order below the “Fractal” patterns low.
3. Place a “target” at previous “swing high.”

Trading Fractal Patterns



Trading Fractal Patterns

The chart above illustrates an example of sell “Fractal” pattern from the Nasdaq futures (NQ). NQ futures rallied in the morning session and closed near 1808. A series of bars attempted to trade “lower lows” and “higher highs” to form a “Fractal” pattern. A trigger bar is anticipated for a short-sell after a “fractal” formation is complete.

1. Wait for trigger bar, which is a “close” below the low of the previous bar to confirm the “fractal” setup.
2. Short below the low of the trigger bar.
3. Place a “stop” order one tick above the high of the trigger bar.
4. Target a major “swing low” prior to the “Fractal” formation.